Model-Based Beam-Steered Optical Wireless Positioning with Single-LED Single-Photodiode for 3D Localization

Kevin Acuna-Condori, Bastien Béchadergue, Hongyu Guan, and Luc Chassagne

Abstract*—State-of-the-art optical wireless positioning (OWP) commonly reaches centimeter-level accuracy by depending on dense multi-light-emitting diodes (LED) infrastructures, photodiode (PD) arrays, or image-sensor receivers, incurring hardware complexity and deployment cost. This paper introduces a single beam-steered LED, single-PD OWP architecture that achieves three-dimensional (3D) localization without receiver rotation, cameras, or PD arrays; the core idea is to steer the transmitter through K known orientations and exploit the resulting received-signal-strength variations at the PD to estimate LED-to-PD direction and distance. We derive a composite Cramér–Rao lower bound and position-error bound (PEB) for the joint observation model, and cast the steering-pattern design as a genetic algorithm that minimizes the PEB over a 3D testbed. We develop both model-based a constrained nonlinear estimator and closed-form direction estimators: a statistically efficient generalized least squares solution, and a lightweight weighted least squares approximation. We derive a direction error bound (DEB) for the direction-finding stage and a position error bound (PEB) for the complete 3D system, and cast the steering-pattern design as a genetic algorithm (GA) that minimizes the DEB over a 3D testbed. We develop three direction estimators: a statistically efficient closed-form generalized least squares (GLS), a lightweight weighted least squares (WLS) approximation, and an iterative nonlinear least-squares (NLS) estimator. We prove that all three estimators are mathematically independent of the receiver orientation, enabling direction finding without receiver pose knowledge or control. Simulations demonstrate centimeter-level accuracy for 3D OWP with a single beam-steered LED and a single PD. Simulations over 1,792 testbed positions with 1,000 Monte Carlo trials demonstrate sub-degree direction-finding accuracy (GLS: 0.66°, NLS: 0.63°, DEB: 0.52°), centimeter-level 3D positioning (GLS: 2.5 cm, NLS: 2.4 cm, PEB: 1.6 cm), and robustness to random receiver tilts with degradation below 3%.

***Index Terms*—3D indoor localization, beam steering, Cramér-Rao bounds, optical wireless communications, optical wireless positioning**

I. Introduction

GLOBAL navigation satellite systems (GNSS) provide ubiquitous outdoor positioning but are unreliable indoors due to severe attenuation, blockage, and multipath. Therefore, there is a need for dedicated indoor position systems (IPS), which use radio-frequency techniques leveraging existing communications infrastructure, such as Wi-Fi [1], Bluetooth Low Energy (BLE) beacons [2], ultra-wideband (UWB) [3], radio-frequency identification (RFID) [4], and emerging 5G positioning [5], are widely studied and deployed [5], [6]. However, in cluttered interiors they are strongly affected by multipath, non-line-of-sight (NLOS) conditions, and time-varying interference in shared spectrum, typically yielding meter-level accuracy unless dense infrastructure, calibration, or specialized hardware is used [7], [8].

In this context, optical wireless positioning (OWP) leverages low-cost, small-form-factor light-emitting diodes (LEDs) that transmit modulated optical signals, often imperceptible to humans, captured by photodiodes (PDs) or cameras to transform lighting into a positioning network [9]. The spatial confinement of light (without wall penetration), immunity to RF electromagnetic interference, high spatial reuse, and more deterministic line-of-sight (LOS) optical channels enable centimeter-level localization with cost-effective luminaires [10]. This makes OWP attractive for navigation and asset tracking in malls, airports, and museums where GNSS is unavailable, as well as in electromagnetic-sensitive environments such as hospitals and airplane cabins [9]. Moreover, the same LED infrastructure can provide optical wireless communication (OWC), often termed light-fidelity (LiFi) [11], so that illumination, data, and positioning coexist in a unified platform. In this work, we deliberately employ a near-infrared (NIR) LED transmitter for positioning, as opposed to conventional visible-light positioning (VLP), which must respect illumination standards and uniformity when reusing luminaires. Since NIR is imperceptible to humans, the positioning link is decoupled from visible-light illumination constraints. This allows us to steer and reorient the source without affecting room lighting [12].

OWP follows two paradigms: data-driven and model-based. Data-driven methods (such as fingerprinting with k-nearest neighbors, support vector machines, random forests, gradient boosting, Gaussian process regression, multilayer perceptrons, convolutional neural networks for structured PD signals, recurrent/temporal models, and transformerTransformer architectures [13], [14]) are effective with abundant labeled data, but these black-box approaches are often opaque, sensitive to domain shift (hardware, illumination, occlusion), and require periodic retraining, offering limited physical guarantees [15]. By contrast, model-based estimators grounded in radiometry and geometry (e.g., Lambertian propagation, field-of-view constraints) provide interpretability, sample efficiency, analyzable uncertainty, and reliable extrapolation, and they support joint system co-design [16].

Most existing OWP implementations rely on traditional multi-LED transmitter constellations, where multiple fixed LEDs with known positions act as anchors [29]. Using multiple LEDs allows the receiver to compute its position via triangulation, trilateration, or multilateration, analogous to

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TABLE I
STATE-OF-THE-ART SINGLE-LED OWP SYSTEMS AND REPORTED PERFORMANCE.

Year	Work	Configuration	Rx Orientation	2D/3D	Method	Error (APE)	Room (m^3)	Notes
2020	[17]	1 LED / 4 PD (1 horizontal + 3 tilted)	Fixed	3D	Model-based (Geometrical equations)	2.52 cm	$1 \times 1 \times 1.5$	Sim.
2021	[18]	1 LED / 3 tilted PD	Fixed	2D	Long Short Term Memory- Fully Connected Network	0.92 cm	$1 \times 1.1 \times 1.75$	Exp.
2021	[19]	1 LED / 4 PD (1 horizontal + 3 tilted)	Fixed	2D	Bayesian algorithm	9.27 cm	$3 \times 3 \times 3$	Exp.
2022	[20]	1 LED / 1 rotatable PD	Fixed	2D	Extreme Learning Machine	1.74 cm	$4 \times 4 \times 3.1$	Sim.
2023	[21]	1 LED / 4 PD	Fixed	2D	Least Square	7 cm ($\text{CDF}_{90\%}$)	$6 \times 6 \times 3$	Exp.
2024	[22]	1 reorientable LED / 1 PD	Arbitrary	2D	Model-based (Power ratio)	5.9 cm ($\text{CDF}_{90\%}$)	$4 \times 4 \times 2.5$	Sim.
2024	[23]	1 LED / 4 PD	Fixed	2D	Extreme Learning Machine / WKNN	2.93 cm	$5 \times 5 \times 3$	Sim.
2024	[24]	1 LED / 4 PD (1 horizontal + 3 tilted)	Fixed	3D	Deep Residual Shrinkage method	5.85 cm	$3.6 \times 3.6 \times 3.0$	Exp.
2024	[25]	1 LED / 1 PD (tilted-rotatable PD)	Rotatable	2D	Least Squares + Improved Grey Wolf Optimizer (LS-IGWO)	1.65 cm	$0.5 \times 0.5 \times 0.3$	Exp.
2024	[26]	1 LED / 4 PD + IMU (orientation)	Arbitrary	3D	MultiLayer Perceptron	1.77 cm	$1.5 \times 1.5 \times 1.5$	Exp.
2025	[27]	1 LED / 1 rotatable PD	Fixed	2D	Least Square	4.96 cm	$5 \times 5 \times 4$	Exp.
2025	[28]	1 LED / 16 IRS / 1 PD	Fixed	2D	Levenberg–Marquardt algorithm	5.6 cm (RMSE)	$4 \times 5 \times 5$	Sim.
2025	Ours	1 reorientable LED / 1 PD	Arbitrary†	3D	Generalized Least Squares	1.54 cm	$3 \times 3 \times 2$	Sim.

Note: "Sim." denotes simulation results; "Exp." denotes experimental results. †Direction finding is provably independent of $\mathbf{n}_r$ (Proposition 1); distance recovery requires a known/controlled $\mathbf{n}_r$.

how GNSS uses many satellites [9]. OWP implementations, specifically VLP, reuses luminaires, and the LED constellation is co-designed with illumination. This co-design includes placement, power, and beam patterns that must satisfy lighting standards and uniformity, which constrains anchor geometry and modulation [30]. However, these multi-LED requirements complicate practical deployment. They presumespresume a dense infrastructure; at least four LEDs must be concurrently visible to the receiver to resolve a three dimensional (3D) position [31]–[34], which may not be true in environments with only sparse lighting. Installing and maintaining a dense LED network, with unique IDs or modulation for each lamp, increases installation cost and complexity [35]. Consequently, reducing the infrastructure burden by localizing with single light sources is a highly desirable goal in the field [13].

Recent research has explored OWP schemes that break from the multi-LED paradigm. One notable direction is single-LED positioning, which aims to achieve localization with only one transmitter [17]–[28]. Table I summarizes representative single-LED OWP systems, reports the average positioning error (APE), and highlights the diversity of receiver designs and estimation methods: many works employ multiple PDs (e.g., 4 PDs or 3 tilted PDs) [17], [19], [21], [23], [24], others rely on a single rotatable PD [20], [25], [27], and some integrate aiding sensors such as an inertial measurement unit [26] or virtualize anchors via intelligent reflecting surfaces (IRS) [28]. Regarding dimensionality, several demonstrations are two dimensional (2D) [18]–[23], [25], [27], [28], whereas fewer achieve 3D with a single LED [17], [24], [26]. A single light source can simplify infrastructure and calibration, but at a fixed orientation one measurement is insufficient to resolve a 3D position. Therefore, single-LED systems typically extract additional information via receiver diversity/rotation, sensor fusion, or engineered propagation (IRS) on the UE side [17], [18], [20], [25], [26], [28]. These limitations motivate the search for a new 3D single-LED single-PD positioning framework.

In parallel, recent advances in OWC have delivered agile beam-steering transmitters. For example, this includes high-speed microelectromechanical systems micromirror gimbals [36], liquid-crystal or metasurface deflectors that widen the field-of-view without sacrificing speed [37], silicon-photonics optical-phased arrays that shift their beam tens of degrees in micro-seconds with no moving parts [38]. In addition, recent OWC/LiFi roadmaps increasingly regard dynamic beam steering as a core enabler for high-capacity, mobile, multiuser links and blockage mitigation making it natural to exploit the same capability for positioning [38]–[40]. These demonstrations confirm that equipping a single indoor luminaire with a steering mechanism is feasible and highly synergistic.

In this work, we study a novel model-based beam–steered single-LED single-PD OWP architecture for 3D indoor localization. The key idea is to steer a single LED through K predefined orientations while a PD at an unknown position collects the resulting K received-signal-strength measurements. The ratio of received powers between any two orientations cancels distance-dependent and receiver-orientation-dependent terms, yielding linear constraints on the transmitter-to-receiver direction. A closed-form eigenvector solution provides the direction estimate. Subsequently, a single cooperative measurement with both the LED and PD aligned along the estimated direction recovers the LED-to-PD distance, completing the 3D position. The main contributions are summarized as follows:

- A novel single-LED single-PD 3D OWP framework that achieves full 3D localization without receiver rotation, camera sensing, or PD arrays, enabled by steering the optical beam of a single LED in K different directions. We show that all proposed direction estimators (GLS, WLS, and NLS) are mathematically independent of the receiver orientation $\mathbf{n}_r$, enabling direction finding without receiver pose knowledge.
- Cramér–Rao lower bound (CRLB) / position error bound (PEB) analysis for a composite observation model. Direction error bound (DEB) and position error bound (PEB) analysis. We derive Fisher-information expressions, define the DEB as the CRLB for the angular direction

error on $\mathbb{S}^2$, and we characterize identifiability and Fisher information matrix (FIM) rank conditions as a function of K.
- Genetic algorithm (GA)-based orientation-set design. We cast the transmitter orientation selection as a GA optimization that minimizes the root-mean-square error (RMSE) of PEB DEB over a 3D testbed, yielding practical design rules for the choice of K and set of orientations.
- Closed-form linear direction estimators. Using a ratio-based linearization, we develop a statistically efficient generalized least squares (GLS) direction estimator, and a lightweight weighted least squares (WLS) approximation, and we evaluate their computational latency.
- An iterative nonlinear least-squares (NLS) direction estimator that fits the normalized Lambertian model on $\mathbb{S}^2$. We show that this estimator is also $\mathbf{n}_r$-independent through power normalization, and achieves direction-finding accuracy closest to the DEB.
- Extensive Monte Carlo simulations (1,000 trials $\times$ 1,792 positions) demonstrate sub-degree direction-finding accuracy, centimeter-level 3D positioning, and robustness to random receiver tilts (degradation $< 3\%$).

This paper is organized as follows: Section II presents the system model and the proposed 3D localization procedure. Section III derives the PEB for single-LED positioning DEB and PEB. Section IV optimizes the transmitter beam-orientation set. Section V develops a nonlinear (NL)the NLS direction estimator. Section VI introduces closed-form linear estimators (GLS and WLS), and establishes receiver-orientation independence. Section VII reports estimators comparison. Section VIII concludes the paper.

Notation: Bold lowercase letters denote column vectors (e.g., $\mathbf{x}$) and bold uppercase letters denote matrices (e.g., $\mathbf{A}$); scalars are in lightface. The Euclidean norm is $\|\cdot\|$, the inner product is $\mathbf{a}\cdot\mathbf{b}$, the trace is $\mathrm{tr}(\cdot)$, and the gradient with respect to $\mathbf{r} = [x, y, z]$ is $\nabla_{\mathbf{r}} = [\partial/\partial x, \partial/\partial y, \partial/\partial z]^{\mathsf{T}}$. The transpose is $(\cdot)^{\mathsf{T}}$, the identity matrix of dimension n and the $n \times m$ zero matrix are $\mathbf{I}_n$ and $\mathbf{0}_{n\times m}$, respectively, and $\mathrm{diag}(\cdot)$ forms a diagonal matrix from its arguments. The smallest eigenvalue and a corresponding unit-norm eigenvector of a symmetric matrix $\mathbf{M}$ are denoted by $\lambda_{\min}(\mathbf{M})$ and $\mathbf{v}_{\min}(\mathbf{M})$; the Loewner order $\succeq$ indicates positive semidefiniteness. $\mathbb{R}^n$ is the n-dimensional real space and $\mathbb{S}^2$ the unit sphere in $\mathbb{R}^3$. Random variables follow the convention $x \sim \mathcal{N}(\mu, \sigma^2)$, with expectation $\mathbb{E}[\cdot]$ and variance $\mathrm{Var}[\cdot]$. A hat $(\hat{\cdot})$ denotes an estimate and an overbar $(\bar{\cdot})$ a sample mean.

II. System Model and Proposed 3D Localization Method

A. System Model

Figure 1 illustrates the single-LED single-PD OWP setup analyzed in this work. A LED transmitter is mounted on the ceiling at a known position $\mathbf{t} = [0, 0, H]^{\mathsf{T}}$ and can steer its optical axis along a unit vector $\mathbf{n}_{t,i} \in \mathbb{S}^2$ for the i-th orientation, $i = 1, \ldots, K$. The receiver is a single PD with fixed normal along the vertical axis normal vector $\mathbf{n}_r = [0, 0, 1]^{\mathsf{T}}$

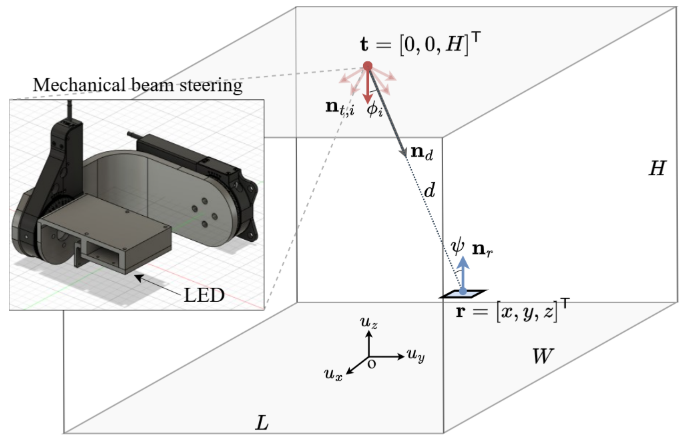

Fig. 1. Single beam-steered LED / single-PD OWP geometry. A beam-steered LED at $\mathbf{t} = [0, 0, H]^{\mathsf{T}}$ (e.g., mechanical steering) points along $\{\mathbf{n}_{t,i}\}_{i=1}^{K}$; the PD is at $\mathbf{r} = [x, y, z]^{\mathsf{T}}$.

$\in \mathbb{S}^2$ located at an unknown position $\mathbf{r} = [x, y, z]^{\mathsf{T}}$. For concreteness, we set $\mathbf{n}_r = [0, 0, 1]^{\mathsf{T}}$ in the simulations. As shown in Section VI-C, all proposed direction estimators are independent of $\mathbf{n}_r$; therefore, the direction-finding results hold for any receiver orientation. Let $\mathbf{d} = \mathbf{r} - \mathbf{t}$ denote the transmitter–receiver displacement vector, $d = \|\mathbf{d}\|$ the separation distance, and $\mathbf{n}_d = \mathbf{d}/d$ the corresponding unit direction vector.

For the i-th LED orientation, the received signal strength (RSS) at the PD is modeled as [41]:

$$S_{r,i} = R_p P_t\, h_{\mathrm{LOS},i} + w_i, \tag{1}$$

where R_p is the PD responsivity, P_t is the transmitted radiant power, and $w_i \sim \mathcal{N}(0, \sigma_w^2)$ models a zero-mean additive white Gaussian noise (AWGN) term arising primarily from shot and Johnson noise, and assumed independent and identically distributed (i.i.d.) across orientations and time (homoscedastic), and $h_{\mathrm{LOS},i}$ denotes the line-of-sight (LOS) channel gain, which for a Lambertian source is given by [41]:

$$h_{\mathrm{LOS},i} = \begin{cases} \dfrac{(m+1)A_{\mathrm{det}}}{2\pi d^2} \cos^m(\phi_i)\cos(\psi), & \psi \le \Psi_{\mathrm{FOV}}, \\ 0, & \text{otherwise}, \end{cases} \tag{2}$$

where m is the Lambertian order of the LED, related to the semi-angle at half power $\Phi_{1/2}$ by $m = -\ln 2/\ln(\cos(\Phi_{1/2}))$, A_{det} the PD effective area, Ψ_{FOV} the receiver field of view, and the irradiance/incidence cosines geometrically given by:

$$\cos\phi_i = \frac{\mathbf{n}_{t,i}\cdot\mathbf{d}}{d}, \qquad \cos\psi = \frac{\mathbf{n}_r\cdot(-\mathbf{d})}{d}. \tag{3}$$

Dividing (1) by R_p yields the received optical powerat a given time instant:

$$P_{r,i} = P_t\, h_{\mathrm{LOS},i} + n_i, \tag{4}$$

where $n_i \sim \mathcal{N}(0, \sigma^2)$ is zero-mean AWGN, remains i.i.d. and homoscedastic, with variance $\sigma^2 = (\sigma_w^2/R_p^2)$.

At every transmitter orientation i, the receiver acquires N independent samples. The sample mean of the received power is:

$$\bar{P}_{r,i} = \frac{1}{N}\sum_{k=1}^{N} P_{r,i,k}, \tag{5}$$

where $P_{r,i,k}$ denotes the k-th instantaneous received power sample at orientation i, corresponding to the k-th realization of the random variable $P_{r,i}$ in (4). Under i.i.d. Gaussian noise, $\bar{P}_{r,i}$ is the minimum-variance unbiased (MVU) estimator of the received power.

The corresponding signal-to-noise ratio (SNR), given the receiver position and a specific transmitter orientation i, noted $\mathrm{SNR}_{r,i}$, as well as the average SNR for a given receiver position, noted SNR_r, and for the entire testbed, noted SNR, are respectively defined as:

$$\mathrm{SNR}_{r,i} = \frac{(R_p P_t\, h_{\mathrm{LOS},i})^2}{\sigma_w^2}, \tag{6a}$$

$$\mathrm{SNR}_{r} = \frac{1}{K}\sum_{i=1}^{K} \mathrm{SNR}_{r,i}, \tag{6b}$$

$$\mathrm{SNR} = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} \mathrm{SNR}_{r}, \tag{6c}$$

where $\mathcal{R}$ denotes the set of receiver testbed positions and $|\mathcal{R}|$ their cardinality.

B. General 3D Single-LED Positioning Procedure

The proposed localization process comprises two consecutive stages:

1) Direction Finding: The first stage estimates the direction vector $\mathbf{n}_d$ by steering the LED through the K predefined orientations $\{\mathbf{n}_{t,i}\}_{i=1}^{K}$ and processing the power means $\{\bar{P}_{r,i}\}_{i=1}^{K}$. Section V (NLNLS estimator) and Section VI (linear GLS/WLS estimators) detail statistically grounded solutions.

2) Distance Recovery: After direction finding, the LED is aligned with the estimated direction $\hat{\mathbf{n}}_d$ and, for beam-forming symmetry, the receiver is oriented to $\mathbf{n}_r = -\hat{\mathbf{n}}_d$. steered to the estimated direction $\hat{\mathbf{n}}_d$ so that its beam is directed toward the receiver, and the PD is reoriented to $\mathbf{n}_r = -\hat{\mathbf{n}}_d$ so that it faces the transmitter. Under this beam-steered configuration and using (2), (3), and (4), the received power for this new orientation $(K+1)$, denoted as $P_{r,K+1}$, obeys:the received optical power at a given time instant for orientation $(K+1)$ obeys:

$$P_{r,K+1} = \frac{C}{d^2} + n_{K+1}, \tag{7}$$

where C is an optical constant calibrated once per hardware set-up:

$$C = \frac{P_t(m+1)A_{\mathrm{det}}}{2\pi}, \tag{8}$$

and n_{K+1} is the noise term for orientation $(K+1)$, following the same statistical properties as the noise described in (4).and $n_{K+1} \sim \mathcal{N}(0, \sigma^2)$ is an i.i.d. noise realization at orientation $(K+1)$, with the same distribution as in (4).

Collecting N independent samples $\{P_{r,K+1,j}\}_{j=1}^{N}$ and using (5), the sample mean satisfies $\bar{P}_{r,K+1} \sim \mathcal{N}\big(C/d^2, \sigma^2/N\big)$. Substituting $\bar{P}_{r,K+1}$ into (7) provides the distance estimate:

$$\hat{d} = \sqrt{\frac{C}{\bar{P}_{r,K+1}}}, \tag{9}$$

and, finally, the 3D estimated position of the receiver, denoted as $\hat{\mathbf{r}}$, is $\hat{\mathbf{r}} = \mathbf{t} + \hat{d}\,\hat{\mathbf{n}}_d$.

This cooperative alignment maximizes both $\cos\phi \approx 1$ and $\cos\psi \approx 1$, thereby maximizing the received power and the SNR of the ranging measurement. The PD reorientation is performed once, after the K direction-finding measurements during which the PD remains at an arbitrary fixed pose. This contrasts with prior single-LED approaches that require continuous PD rotation [20], [25], [27].

III. Position Error BoundDirection Error Bound and Position Error Bound

This section derives the Cramér–Rao lower bound (CRLB) for direction estimation from the K steered-orientation measurements. We term this limit the direction error bound (DEB). As an extension, we also define the position error bound (PEB), which additionally accounts for the distance-recovery measurement and serves as a benchmark for the full 3D pipeline.

A. Joint Statistical Model of the Observations

According to Section II, the sample-mean optical power, as defined in (5), for each of the K transmitter orientations used during the direction finding stage is distributed as:

$$\bar{P}_{r,i} \sim \mathcal{N}\big(\mu_i(\mathbf{r}),\, \sigma^2/N\big), \quad i = 1,\ldots,K, \tag{10}$$

where $\mu_i(\mathbf{r}) = P_t\, h_{\mathrm{LOS},i}(\mathbf{r})$ and $h_{\mathrm{LOS},i}$ is defined in (2) with the cosines in (3). During the distance recovery stage, the mean optical power in aligned configuration obeys:

$$\bar{P}_{r,K+1} \sim \mathcal{N}\big(\mu_{K+1}(\mathbf{r}),\, \sigma^2/N\big), \tag{11}$$

where $\mu_{K+1}(\mathbf{r}) = C/d^2(\mathbf{r})$, $d(\mathbf{r}) = \|\mathbf{r} - \mathbf{t}\|$ and C is the optical constant introduced in (8).

Stacking these $K+1$ independent observations into $\mathbf{z} = [\,\bar{P}_{r,1}, \ldots, \bar{P}_{r,K}, \bar{P}_{r,K+1}\,]^{\mathsf{T}}$ yields the multivariate Gaussian model:

$$\mathbf{z} = \{z_i\}_{i=1,\cdots,K+1} \sim \mathcal{N}\big(\boldsymbol{\mu}(\mathbf{r}),\, (\sigma^2/N)\,\mathbf{I}_{K+1}\big), \tag{12}$$

with mean vector $\boldsymbol{\mu}(\mathbf{r}) = [\,\mu_1(\mathbf{r}), \ldots, \mu_{K+1}(\mathbf{r})\,]^{\mathsf{T}}$ and diagonal covariance σ^2/N.

B. Log-Likelihood and Score Function

Under the assumption of independent Gaussian measurements, the log-likelihood of $\mathbf{r}$ given $\mathbf{z}$ is:

$$\ell(\mathbf{r}) = \ln L(\mathbf{r} \mid \mathbf{z}) = -\frac{N}{2\sigma^2}\sum_{i=1}^{K+1}\big[z_i - \mu_i(\mathbf{r})\big]^2 + c_0, \tag{13}$$

where c_0 is an additive constant independent of $\mathbf{r}$ [42]. Differentiating $\ell(\mathbf{r})$ with respect to $\mathbf{r}$ defines the score vector:

$$\mathbf{s}(\mathbf{r}) = \nabla_{\mathbf{r}}\, \ell(\mathbf{r}) = \frac{N}{\sigma^2} \sum_{i=1}^{K+1} \big[z_i - \mu_i(\mathbf{r})\big]\, \nabla_{\mathbf{r}}\, \mu_i(\mathbf{r}), \tag{14}$$

where $\nabla_{\mathbf{r}} = [\partial/\partial x, \partial/\partial y, \partial/\partial z]^{\mathsf{T}}$ is the gradient operator.

C. Fisher Information Matrix

For distributions in the exponential family, such as the homoscedastic Gaussian model assumed here, the FIM can be obtained as the variance of the score vector, avoiding second-order differentiation and yielding greater numerical stability [43]:

$$\mathcal{I}(\mathbf{r}) = \mathrm{Var}\big[\mathbf{s}(\mathbf{r})\big] = \mathbb{E}\big[\mathbf{s}(\mathbf{r})\, \mathbf{s}(\mathbf{r})^{\mathsf{T}}\big] \tag{15}$$

$$= \frac{N}{\sigma^2} \sum_{i=1}^{K+1} \big[\nabla_{\mathbf{r}}\mu_i(\mathbf{r})\big]\, \big[\nabla_{\mathbf{r}}\mu_i(\mathbf{r})\big]^{\mathsf{T}}. \tag{16}$$

The second equality follows from $\mathbb{E}[z_i] = \mu_i(\mathbf{r})$, which implies $\mathbb{E}\big[(z_i - \mu_i)(z_j - \mu_j)\big] = 0$ for $i \neq j$. Hence the total information is the coherent superposition of the differential contributions from each LED orientation and the range measurement. Required gradients are:

1) Direction-Finding Measurements: For the i-th orientation:

$$\nabla_{\mathbf{r}}\mu_i(\mathbf{r}) = P_t\, \nabla_{\mathbf{r}} h_{\mathrm{LOS},i}(\mathbf{r}), \tag{17}$$

which, by differentiating (2) via the chain rule on $d(\mathbf{r})$, $\cos\phi_i$ and $\cos\psi$, yields the closed-form:

$$\nabla_{\mathbf{r}}\mu_i(\mathbf{r}) = \frac{C}{d^3}\Big[m\, \cos^{m-1}\phi_i\, \cos\psi\, \mathbf{n}_{t,i} - \cos^m\phi_i\, \mathbf{n}_r - (m+3)\, \cos^m\phi_i\, \cos\psi\, \mathbf{n}_d\Big], \tag{18}$$

which vanishes whenever $\psi > \Psi_{\mathrm{FOV}}$ since $h_{\mathrm{LOS},i} = 0$ outside the receiver's field of view.

2) Distance-Recovery Measurement:

$$\nabla_{\mathbf{r}}\mu_{K+1}(\mathbf{r}) = -\frac{2C}{d^4}(\mathbf{r} - \mathbf{t}) = -\frac{2C}{d^3}\, \mathbf{n}_d, \tag{19}$$

where $\mathbf{n}_d$ is the unit direction vector defined in Section II.

D. Formal Definition of the PEB

Let $\hat{\mathbf{r}}$ be any unbiased estimator of $\mathbf{r}$. The Cramér–Rao inequality asserts: $\mathrm{Cov}[\hat{\mathbf{r}}] \succeq \mathcal{I}^{-1}(\mathbf{r})$. Accordingly, a scalar performance metric, the PEB, is defined as:

$$\mathrm{PEB}(\mathbf{r}) = \sqrt{\mathrm{tr}\big\{\mathcal{I}^{-1}(\mathbf{r})\big\}}. \tag{20}$$

The square root of the trace of $\mathcal{I}^{-1}$ corresponds to the minimum achievable RMSE in position estimation for the described OWP configuration. A small PEB indicates that the chosen set of LED orientations and the range measurement provide sufficient and complementary information for high-precision localization; conversely, a large PEB suggests the need to increase K, improve the SNR, or redesign the orientation pattern.

E. Direction-Finding Observation Model

At each transmitter orientation $i = 1, \ldots, K$, the sample-mean received power (5) is distributed as:

$$\bar{P}_{r,i} \sim \mathcal{N}\big(\mu_i(\mathbf{r}),\, \sigma^2/N\big), \quad i = 1, \ldots, K. \tag{21}$$

From (4)–(3), the noise-free mean power factorizes as:

$$\mu_i = \eta\, Q_i^m, \tag{22}$$

where:

$$Q_i = \mathbf{n}_{t,i} \cdot \mathbf{n}_d = \cos\phi_i \tag{23}$$

is the direction cosine between the i-th LED orientation and the true transmitter-to-receiver direction $\mathbf{n}_d$, and

$$\eta = \frac{C\, \cos\psi}{d^2} \tag{24}$$

is a positive scalar common to all K orientations. The constant η absorbs the unknown distance d and the receiver orientation $\mathbf{n}_r$ (through $\cos\psi$). Since the direction-finding stage aims to recover $\mathbf{n}_d$ alone, η is a nuisance parameter that must be accounted for in the bound derivation.

F. Fisher Information for Direction Estimation

We parameterize the direction in nadir-referenced spherical coordinates:

$$\mathbf{n}_d = \big[-\sin\theta_d \cos\phi_d,\ -\sin\theta_d \sin\phi_d,\ -\cos\theta_d\big]^{\mathsf{T}}, \tag{25}$$

where $\theta_d \in [0, \pi)$ is the polar angle from $-\mathbf{u}_z$ and $\phi_d \in [0, 2\pi)$ is the azimuth. Let $\mathbf{u}_\theta = \partial\mathbf{n}_d/\partial\theta_d$ and $\mathbf{u}_\phi = \partial\mathbf{n}_d/\partial\phi_d$ denote the tangent vectors to $\mathbb{S}^2$ at $\mathbf{n}_d$.

We adopt $\alpha \triangleq \ln\eta$ as the nuisance coordinate (the log-reparameterization ensures positivity and improves numerical conditioning) and form the extended parameter vector $\boldsymbol{\xi} = [\theta_d,\, \phi_d,\, \alpha]^{\mathsf{T}} \in \mathbb{R}^3$. Under i.i.d. Gaussian noise with variance σ^2/N, the Slepian–Bangs formula [43] yields the 3×3 FIM:

$$\mathcal{I}_{\mathrm{DF}}(\boldsymbol{\xi}) = \frac{N\,\eta^2}{\sigma^2} \sum_{i=1}^{K} \tilde{\mathbf{g}}_i\, \tilde{\mathbf{g}}_i^{\mathsf{T}}, \tag{26}$$

where the normalized gradient vector for the i-th orientation is:

$$\tilde{\mathbf{g}}_i = \big[m\, Q_i^{m-1}(\mathbf{n}_{t,i} \cdot \mathbf{u}_\theta),\ \ m\, Q_i^{m-1}(\mathbf{n}_{t,i} \cdot \mathbf{u}_\phi),\ \ Q_i^m\big]^{\mathsf{T}}. \tag{27}$$

The first two components capture the sensitivity of μ_i to angular perturbations of $\mathbf{n}_d$, while the third component captures sensitivity to the amplitude η.

G. Direction Error Bound (DEB)

Since α is not of interest, it is profiled out via the Schur complement of the $(3,3)$ block of $\mathcal{I}_{\mathrm{DF}}$ [43]. The resulting CRLB for the angular parameters (θ_d, ϕ_d) is the upper-left 2×2 block of the inverse FIM:

$$\mathbf{C}_{\mathrm{ang}} = \big[\mathcal{I}_{\mathrm{DF}}^{-1}(\boldsymbol{\xi})\big]_{1:2,\,1:2}. \tag{28}$$

This matrix provides a tight lower bound on the covariance of any unbiased estimator of (θ_d, ϕ_d), irrespective of how the nuisance parameter η is handled (estimated jointly, profiled, or known).

To express the bound in terms of the Cartesian direction error $\|\hat{\mathbf{n}}_d - \mathbf{n}_d\|$, we project through the Jacobian $\mathbf{J}_{\text{sph}} = [\mathbf{u}_\theta,\, \mathbf{u}_\phi] \in \mathbb{R}^{3\times 2}$:

$$\text{DEB}(\mathbf{r}) = \sqrt{\text{tr}\big\{\mathbf{J}_{\text{sph}}\, \mathbf{C}_{\text{ang}}\, \mathbf{J}_{\text{sph}}^{\mathsf{T}}\big\}}. \tag{29}$$

For small errors, DEB $\approx$ angular RMSE in radians. The DEB depends on the receiver position $\mathbf{r}$ (through η and $\{Q_i\}$), on the orientation set $\{\mathbf{n}_{t,i}\}$, and on the system parameters $(N, \sigma^2, m, P_t, A_{\text{det}})$. Crucially, it uses only the K direction-finding measurements and is therefore the appropriate benchmark for direction estimators.

H. Position Error Bound (PEB)

When the full two-stage pipeline is evaluated (direction finding followed by distance recovery), an additional measurement $\bar{P}_{r,K+1} \sim \mathcal{N}(C/d^2, \sigma^2/N)$ becomes available. Including this $(K+1)$-th observation in the FIM yields the Cartesian-parameterized 3×3 matrix:

$$\mathcal{I}(\mathbf{r}) = \frac{N}{\sigma^2} \sum_{i=1}^{K+1} \big[\nabla_{\mathbf{r}}\mu_i(\mathbf{r})\big]\, \big[\nabla_{\mathbf{r}}\mu_i(\mathbf{r})\big]^{\mathsf{T}}, \tag{30}$$

where $\nabla_{\mathbf{r}}\mu_i$ for $i = 1, \ldots, K$ is given by

$$\nabla_{\mathbf{r}}\mu_i(\mathbf{r}) = \frac{C}{d^3}\Big[m\, \cos^{m-1}\phi_i\, \cos\psi\, \mathbf{n}_{t,i} - \cos^m\phi_i\, \mathbf{n}_r - (m+3)\, \cos^m\phi_i\, \cos\psi\, \mathbf{n}_d\Big], \tag{31}$$

and the distance-recovery gradient is $\nabla_{\mathbf{r}}\mu_{K+1} = -2C\,\mathbf{n}_d/d^3$. The PEB is defined as:

$$\text{PEB}(\mathbf{r}) = \sqrt{\text{tr}\big\{\mathcal{I}^{-1}(\mathbf{r})\big\}}. \tag{32}$$

Since the PEB assumes perfect beam alignment in the $(K+1)$-th measurement, it is an optimistic lower bound that no practical two-stage estimator can attain [43]. In this work, the DEB (29) is adopted as the primary metric and GA objective; the PEB is reported in Section VII for completeness.

I. Number of LED Orientations

1) Underdetermined Regime ($K \le 2$): In this regime, the Fisher information matrix $\mathcal{I}(\mathbf{r})$ from (30) is rank-deficient. Each of the K orientation measurements contributes at most to one independent gradient vector, and the distance-recovery measurement adds a single additional vector. When $K \le 2$, there are fewer than three linearly independent gradients in total, so $\mathcal{I}(\mathbf{r})$ is singular and the CRLB on any unbiased position estimate diverges [29].

2) Just-Determined Regime ($K = 3$): With three distinct, non-coplanar LED orientations plus the range measurement, the total of four gradient vectors can span the 3D parameter space. In generic geometries the FIM is full rank and $\mathcal{I}^{-1}(\mathbf{r})$ exists, yielding a finite PEB. However, the absence of redundancy makes the bound highly sensitive to small perturbations in orientation geometry [18].

3) Overdetermined Regime ($K \ge 4$): For four or more orientations, the FIM accumulates redundant outer- products of gradients. This overdetermination improves the numerical conditioning of $\mathcal{I}(\mathbf{r})$ and generally lowers the PEB, since extra measurements provide additional independent information beyond the minimal requirement [9], [26].

IV. Optimization of the Orientation Set

Equation (31) shows that the FIM $\mathcal{I}(\mathbf{r})$ depends explicitly on the transmitter orientation vectors $\{\mathbf{n}_{t,i}\}_{i=1}^{K}$, and hence so does the PEB in (32). We thus define the objective function as the RMSE of the PEB values evaluated over all receiver positions within the 3D testbed specified in Table II. The DEB in (29) depends explicitly on the transmitter orientation set $\{\mathbf{n}_{t,i}\}_{i=1}^{K}$ through the direction-finding FIM $\mathcal{I}_{\text{DF}}(\boldsymbol{\xi})$ in (26). We define the objective function as the RMSE of the DEB evaluated over all receiver positions in the 3D testbed specified in Table II:

$$f(\mathbf{x}) = \sqrt{\frac{1}{|\mathcal{R}|}\sum_{\mathbf{r}\in\mathcal{R}} \text{DEB}^2(\mathbf{r};\, \mathbf{x})}, \tag{33}$$

where $\mathbf{x} = [\theta_1, \varphi_1, \ldots, \theta_K, \varphi_K]^{\mathsf{T}}$ encodes the orientation set. Selecting the orientation set that minimizes this objective is essential for localization direction-finding accuracy. However, PEBDEB($\{\mathbf{n}_{t,i}\}$) is nonlinear in the orientation parameters, involves multiple coupled terms (e.g., powers of cosines and distance-dependent factors), and exhibits nonsmooth behavior due to the receiver FOV. Consequently, closed-form optimization is not available and purely local methods are prone to suboptimal solutions. To robustly explore the parameter space and approach a globally optimal orientation set for a given K, we employ a GA [44].

TABLE II
PARAMETERS USED FOR GA OPTIMIZATION.

System Parameters	
Parameter	**Value**
Transmitter coordinates ($\mathbf{t}$)	$[0, 0, 2]^{\mathsf{T}}$ m
Transmitted optical power (P_t)	0.405 W
Half-power semi-angle ($\Phi_{1/2}$)	$\{30°, 45°, 60°\}$
Lambertian order (m)	$-\ln 2/\ln(\cos(\Phi_{1/2}))$
Photodiode effective area (A_{det})	$4.8 \times 5.5\,\text{mm}^2$
Receiver field of view (Ψ_{FOV})	85°
Electrical noise variance (σ_w^2)	$1.19 \times 10^{-14}\,\text{A}^2$
PD responsivity (R_p)	0.63 A/W
Noise variance (σ^2)	$3 \times 10^{-14}\,\text{W}^2$
Samples per orientation (N)	1000
Genetic Algorithm	
Parameter	**Value**
Population size (P)	300
Maximum generations ($G_{\max}$)	200
Crossover probability (p_c)	0.8
Search range (Ω)	$\theta_i \in [0, 80°]$, $\varphi_i \in [0, 360°]$
Objective function ($f(\mathbf{x})$)	RMSE of the PEBRMSE of the DEB (33)
Decision variables ($\mathbf{x}$)	$[\theta_1, \varphi_1, \ldots, \theta_K, \varphi_K]^{\mathsf{T}}$
3D Testbed	
Parameter	**Value**
Room dimensions ($L \times W \times H$)	$3 \times 3 \times 2\,\text{m}^3$
Range in x, y	$[-1.5,\ 1.5]$ m
Range in z	$[0,\ 1.2]$ m
Grid step	0.2 m
Total number of points tested $\lvert\mathcal{R}\rvert$	1792

TABLE III
DEB-OPTIMIZED ORIENTATIONS (θ_i, φ_i) IN DEGREES $[°]$ FOR VARIOUS K IN A $3 \times 3 \times 2\,\mathrm{m}^3$ ROOM, WITH THE ACHIEVED RMS-DEB.

K	**RMS-DEB**	$i=1$	$i=2$	$i=3$	$i=4$	$i=5$	$i=6$	$i=7$	$i=8$	$i=9$
3	1.30°	(17.5, 203.7)	(17.2, 332.2)	(18.7, 88.8)	-	-	-	-	-	-
4	0.75°	(29.9, 315.0)	(29.9, 135.0)	(29.9, 45.0)	(29.9, 225.0)	-	-	-	-	-
5	0.52°	(0.1, 74.2)	(65.7, 269.8)	(65.7, 179.9)	(65.9, 359.8)	(65.9, 89.9)	-	-	-	-
6	0.47°	(67.6, 253.4)	(66.6, 321.1)	(70.0, 176.9)	(0.1, 274.3)	(68.9, 98.4)	(66.8, 24.9)	-	-	-
7	0.41°	(65.6, 353.3)	(2.5, 10.1)	(66.2, 273.6)	(64.6, 196.5)	(62.5, 130.8)	(2.6, 191.7)	(63.8, 68.1)	-	-
8	0.39°	(67.6, 67.1)	(66.6, 247.3)	(2.8, 39.7)	(3.2, 227.0)	(64.7, 2.8)	(66.1, 179.7)	(66.9, 298.2)	(65.4, 114.9)	-
9	0.36°	(66.1, 165.4)	(8.7, 267.1)	(66.8, 273.8)	(62.4, 219.4)	(64.2, 21.6)	(67.2, 91.0)	(13.9, 105.1)	(4.5, 303.1)	(62.0, 330.3)

Within this GA framework, each individual encodes K steering directions via tilt–azimuth pairs $\{(\theta_i, \varphi_i)\}_{i=1}^{K}$, yielding a $2K$-dimensional genotype $\mathbf{x} = [\theta_1, \varphi_1, \ldots, \theta_K, \varphi_K]^\mathsf{T}$. We adopt a nadir-referenced spherical parameterisation where $\mathbf{u}_x = [1,0,0]^\mathsf{T}$, $\mathbf{u}_y = [0,1,0]^\mathsf{T}$, $\mathbf{u}_z = [0,0,1]^\mathsf{T}$ are the standard Cartesian basis vectors, θ_i is the tilt from $-\mathbf{u}_z$ and φ_i is the azimuth in the $\mathbf{u}_x$–$\mathbf{u}_y$ plane measured counterclockwise from $+\mathbf{u}_x$, with domains $\theta_i \in [0, \theta_{\max}]$ and $\varphi_i \in [0°, 360°]$. The mapping to unit-norm orientation vectors in the reference frame $(\mathbf{u}_x, \mathbf{u}_y, \mathbf{u}_z)$ is:

$$\mathbf{n}_{t,i} = \left[\sin\theta_i \cos\varphi_i,\ \sin\theta_i \sin\varphi_i,\ -\cos\theta_i\right]^\mathsf{T}. \tag{34}$$

Spherical coordinates are preferred to Cartesian because they describe unit-norm directions with two angles, avoiding the nonlinear unit-norm constraint and providing bounded domains that simplify GA search. Standard genetic operators—tournament selection, crossover with probability p_c, and adaptive feasible mutation—are applied for at most $G_{\max}$ generations, with early termination when the average relative improvement of the best fitness over $G_{\max}$ generations falls below 10^{-6}. The implementation is in MATLAB; optimization parameters are summarized in Table II.

We emphasize that the GA optimization is an offline system-design problem, not an online estimation problem. The $2K$ decision variables (tilt–azimuth pairs) define the orientation set; for each candidate set, the DEB (or PEB) is evaluated analytically over the entire 3D testbed using the closed-form FIM in (30). No measurements are involved in this optimization—only theoretical performance bounds. The orientation set is optimized to minimize the RMSE of the DEB averaged over all $|\mathcal{R}| = 1{,}792$ receiver positions (Table II). The position $\mathbf{r}$ is therefore not fixed but sweeps the spatial grid during fitness evaluation; the resulting orientation set is position-independent and can be deployed universally. Once fixed, the online direction-finding stage uses K power measurements to recover the 2-DoF direction on $\mathbb{S}^2$, which is overdetermined for $K \geq 3$.

Table III lists the DEB-optimal K-orientation sets for the system parameters in Table II, together with the achieved RMS-DEB. Each set is unique and no set is a subset of another. The RMS-DEB decreases monotonically from 1.30° at $K = 3$ to 0.36° at $K = 9$, showing the benefit of additional orientations for direction-finding accuracy. At the selected operating point $K = 5$, the DEB-optimized set achieves an RMS-DEB of 0.52°.

In Fig. 2, for each K we present box-and-whisker plots of the PEB DEB computed over all receiver locations in the $3 \times 3 \times 2\,\mathrm{m}^3$ testbed; optimized sets are shown in green, randomly selected sets in orange, and the shaded profiles depict the corresponding probability density function (PDF) estimates. Quantitatively, the optimized sets achieve a median of PEB below 1.5 cm for all $K \in \{3, \ldots, 9\}$ (worst median 1.43 cm at $K = 3$) and reach 0.74 cm at $K = 9$. The GA consistently outperforms random orientations: across K, the optimized sets reduce the median PEB DEB by 49%–72% and exhibit smaller interquartile ranges. These gains are reflected in the distributions, with the central tendency improving monotonically as K increases.

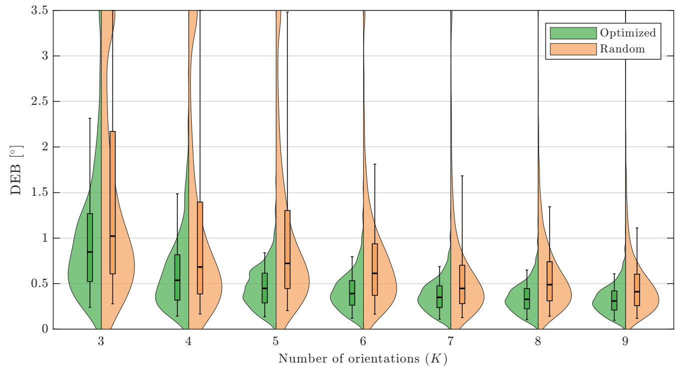

Fig. 2. Distribution of the PEB across all receiver locations in the $3\times3\times2\,\mathrm{m}^3$ testbed. For each K, green shows the GA-optimized orientation set and orange a randomly selected set. Boxes and whiskers summarize the sample distribution, while the shaded profiles depict the corresponding PDF.Distribution of the DEB [°] across all receiver locations in the $3 \times 3 \times 2\,\mathrm{m}^3$ testbed for $K \in \{3, \ldots, 9\}$. Green: GA-optimized orientation set; orange: randomly selected set. Box-and-whisker plots summarize the sample distribution; shaded profiles depict the estimated PDF. The GA-optimized sets consistently achieve lower DEB and tighter distributions across all K.

Figure 3 presents spatial heat maps of the PEBDEB across a $3 \times 3\,\mathrm{m}^2$ floor at $z = 0.8$ m for $K = 5$ orientations with the transmitter at the room centre. In the optimal configuration, shown in Fig. 3(a), the highest errors occur uniformly along the boundaries and especially at the corners, analogous to multi-transmitter layouts [45], but remain spatially consistent. By contrast, the random configuration in Fig. 3(b) exhibits both larger and more irregular error patterns.

We now turn from optimization to performance characterization: using the optimized orientation sets, we quantify how the 90th-percentile of the PEB distribution ($\mathrm{PEB}_{90\%}$) varies with K, $\Phi_{1/2}$, and SNR. Figure 4 depicts the $\mathrm{PEB}_{90\%}$ for the optimal K-orientation set under three $\Phi_{1/2}$ scenarios at an effective SNR (from (6c)) of 14 dB, selected as a representative moderate-SNR setting and consistent with prior indoor VLP evaluations that analyze performance around 15 dB [46]. In all scenarios, $\mathrm{PEB}_{90\%}$ decreases as K increases,

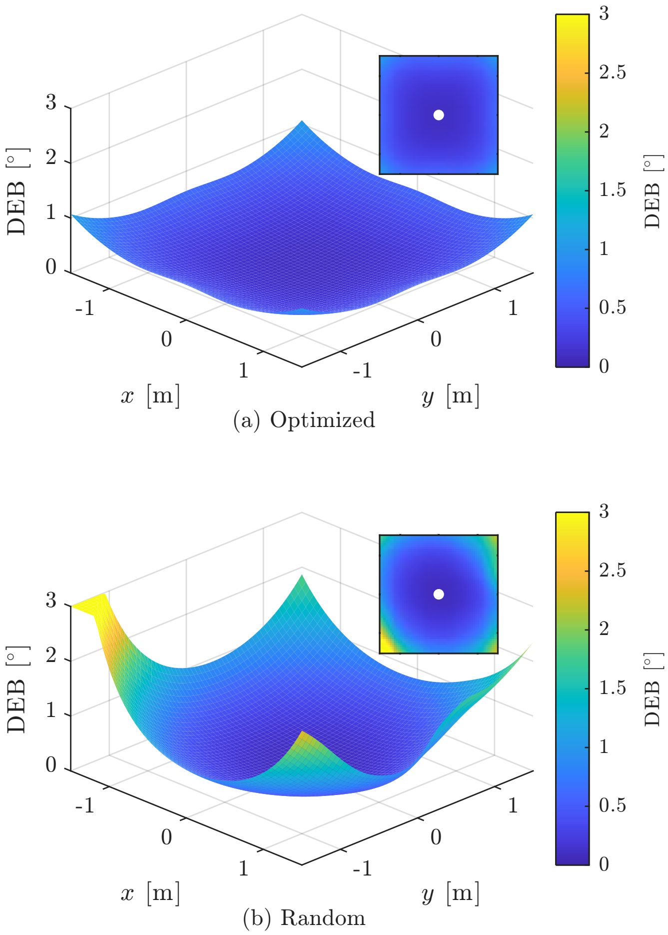

Fig. 3. Heat map of the PEB across a $3 \times 3\,\mathrm{m}^2$ testbed at $z = 0.8\,\mathrm{m}$ for $K = 5$ orientations: (a) optimal set, (b) random set.Spatial distribution of the DEB across the $3 \times 3\,\mathrm{m}^2$ floor at $z = 0.8\,\mathrm{m}$ for $K = 5$ orientations. (a) GA-optimized set; (b) randomly selected set. The colour scale is shared and clipped at $3°$.

indicating improved localization precision with additional orientations. Assuming a fixed per-orientation acquisition time, the total acquisition latency scales linearly with K; under this accuracy–latency tradeoff, $K = 5$ offers a favorable compromise. For $\Phi_{1/2} = 30°$, at least four orientations are necessary due to the increased directivity limiting coverage with only three orientations; configurations with $K \geq 4$ outperform alternatives. In contrast, $\Phi_{1/2} = 45°$ achieves competitive error performance starting at $K = 3$, and given its widespread use in commercial LEDs, it is adopted for the subsequent simulations. We now turn from optimization to performance characterization. Figure 4 plots the RMS-DEB (solid, left axis) and RMS-PEB (dashed, right axis) versus K for the DEB-optimized orientation sets under three half-power angles $\Phi_{1/2} \in \{45°, 60°, 75°\}$. Both bounds decrease monotonically with K, confirming that additional orientations improve direction-finding and positioning accuracy simultaneously. The direction-finding bound reaches sub-degree RMS-DEB at $K \geq 5$ for $\Phi_{1/2} = 45°$, while the corresponding RMS-PEB falls below $2\,\mathrm{cm}$. Wider beams ($\Phi_{1/2} = 60°$–$75°$) incur higher bounds due to the reduced Lambertian directivity, yet the trends remain consistent. Assuming a fixed per-orientation acquisition time, latency scales linearly with K; under this accuracy–latency trade-off, $K = 5$ offers a favourable compromise for both bounds. $\Phi_{1/2} = 45°$ achieves competitive performance starting at $K = 3$ and, given its widespread use in commercial LEDs, is adopted for the subsequent simulations.

Figure 5 presents the RMS-DEB (solid, left axis) and RMS-PEB (dashed, right axis) versus SNR for the DEB-optimized orientation sets with $\Phi_{1/2} = 45°$. The bold curves correspond to the recommended operating point $K = 5$; the shaded bands span the range $K \in \{3, \dots, 9\}$. Both bounds decrease monotonically with SNR and exhibit nearly identical slopes of $-1/2$ on the log–log axes, confirming the theoretical $1/\sqrt{\mathrm{SNR}}$ scaling of the CRLB under Gaussian noise [43]. At a representative SNR of $14\,\mathrm{dB}$, the $K = 5$ set achieves an RMS-DEB below $0.3°$ and an RMS-PEB below $1\,\mathrm{cm}$. The band width at any fixed SNR quantifies the gain from additional orientations; it narrows at high SNR, indicating diminishing returns from increasing K in low-noise regimes.

V. Nonlinear Estimator Nonlinear Least-Squares Direction Estimator

The optical power collected by the PD depends nonlinearly on its relative position to the LED, through the distance d, the irradiance angle ϕ_i, and the incidence angle ψ. Denoting the orientation vector of the receiver and the transmitter as $\mathbf{n}_r = [\alpha, \beta, \gamma]^\mathsf{T}$ and $\mathbf{n}_{t,i} = [a_i, b_i, c_i]^\mathsf{T}$, respectively, from (3), we have:

This section develops an iterative nonlinear least-squares (NLS) direction estimator that operates directly on the Lambertian power model without the linearization employed by GLS/WLS (Section VI). The key idea is to normalize the received powers so as to eliminate the common multiplicative factor that depends on the unknown distance d and receiver orientation $\mathbf{n}_r$, thereby rendering the estimator $\mathbf{n}_r$-independent.

A. NLS Cost Function and Direction Estimate

As established in Section III-F, the noise-free received power factorizes as $\mu_i = \eta\, Q_i^m$ (22), where $Q_i = \mathbf{n}_{t,i} \cdot \mathbf{n}_d$ (23) and $\eta = C \cos\psi / d^2$ (24) is a common positive scalar that absorbs the unknown distance and receiver orientation.

Let $\hat{\mu}_i = \bar{P}_{r,i}$ denote the sample-mean power at orientation i (cf. (5)). We define the normalized power targets as

$$p_i \triangleq \frac{\hat{\mu}_i}{\max_{j=1,\dots,K} \hat{\mu}_j}, \quad i = 1, \dots, K. \tag{35}$$

In the noiseless regime, substituting (22) yields

$$p_i = \frac{\eta\, Q_i^m}{\eta\, Q_{\max}^m} = \frac{Q_i^m}{Q_{\max}^m}, \tag{36}$$

where $Q_{\max} = \max_j Q_j$. The common factor η cancels exactly, so the normalized targets p_i depend only on the LED orientations $\{\mathbf{n}_{t,i}\}$ and the direction $\mathbf{n}_d$, not on $\mathbf{n}_r$, d, P_t, or A_{det}.

We model the normalized targets as $p_i \approx \eta\, [\max(0, \mathbf{n}_{t,i} \cdot \mathbf{v})]^m$, where $\mathbf{v} \in \mathbb{S}^2$ is the candidate direction and $\eta > 0$ is a

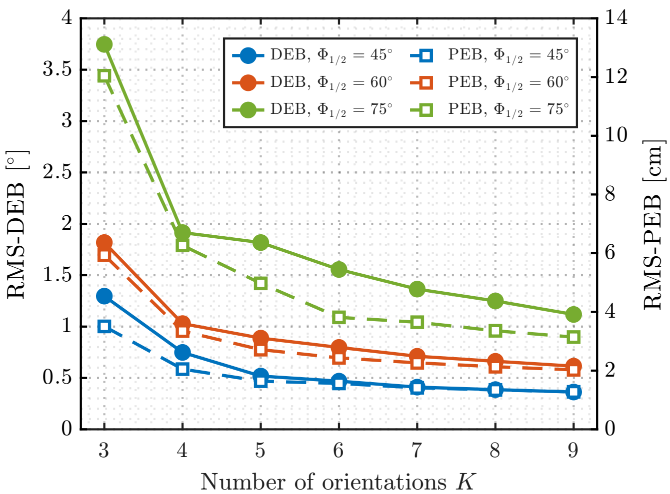

Fig. 4. $\text{PEB}_{90\%}$ versus the number of orientations K for GA-optimized sets at a fixed SNR of 14 dB and $\Phi_{1/2} \in \{30°, 45°, 60°\}$. Values are computed over all receiver positions in the $3 \times 3 \times 2$ m^3 testbed.RMS-DEB (solid, left axis) and RMS-PEB (dashed, right axis) versus the number of orientations K for the DEB-optimized sets and three LED half-power angles $\Phi_{1/2} \in \{45°, 60°, 75°\}$. Values are computed over all receiver positions in the $3 \times 3 \times 2$ m^3 testbed. Both bounds decrease monotonically with K; the operating point $K = 5$ offers a favourable accuracy–latency trade-off for both direction finding and 3D positioning.

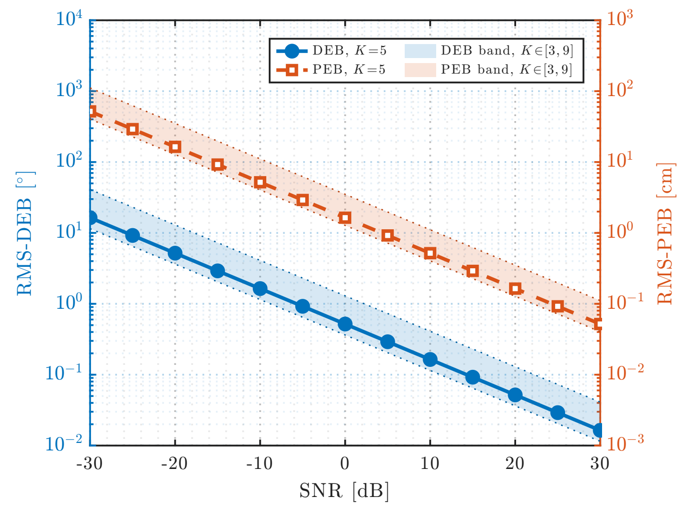

Fig. 5. $\text{PEB}_{90\%}$ versus SNR for GA-optimized orientation sets with $K \in \{3, \ldots, 9\}$ ($\Phi_{1/2} = 45°$; remaining parameters in Table II). Values are computed over all receiver positions in the $3 \times 3 \times 2\,\text{m}^3$ testbed.RMS-DEB (solid, left axis) and RMS-PEB (dashed, right axis) versus SNR for the DEB-optimized orientation sets ($\Phi_{1/2} = 45°$). Bold curves: $K = 5$; shaded bands: range over $K \in \{3, \ldots, 9\}$; dotted edges: $K = 3$ (upper) and $K = 9$ (lower). Both bounds decrease as $1/\sqrt{\text{SNR}}$, with the band width reflecting the diminishing incremental gain from additional orientations at high SNR.

free scale parameter that absorbs the unknown normalization constant $1/Q_{\max}^m$. The NLS direction estimator is then

$$(\hat{\mathbf{n}}_{d,\text{NLS}},\, \hat{\eta}) = \arg \min_{\substack{\mathbf{v} \in \mathbb{S}^2,\ \eta > 0 \\ \mathbf{n}_{t,i} \cdot \mathbf{v} \geq 0,\ i=1,\ldots,K}} F(\mathbf{v}, \eta), \tag{37}$$

with cost function

$$F(\mathbf{v}, \eta) = \sum_{i=1}^{K} \big(\eta\, [\mathbf{n}_{t,i} \cdot \mathbf{v}]^m - p_i\big)^2. \tag{38}$$

The unit-sphere constraint $\|\mathbf{v}\| = 1$ restricts the search to directions, while the positivity constraints $\mathbf{n}_{t,i} \cdot \mathbf{v} \geq 0$ enforce that all K orientations illuminate the receiver (i.e., $\cos\phi_i \geq 0$).

B. Consistency and Optimality at the True Direction

At the true direction $\mathbf{v} = \mathbf{n}_d$, we have $\mathbf{n}_{t,i} \cdot \mathbf{v} = Q_i$ and hence $[\mathbf{n}_{t,i} \cdot \mathbf{v}]^m = Q_i^m$. From (36), $p_i = Q_i^m / Q_{\max}^m$. Substituting into (38):

$$F(\mathbf{n}_d, \eta) = \sum_{i=1}^{K} Q_i^{2m} \big(\eta - 1/Q_{\max}^m\big)^2 = 0 \tag{39}$$

when $\eta^* = 1/Q_{\max}^m$. Thus, in the absence of noise, the NLS achieves a global minimum of zero at the true direction, confirming consistency.

Since $\mathbf{n}_r$ does not appear in (35)–(38), the NLS direction estimate is independent of the receiver orientation by construction. The common factor η is eliminated through max-normalization and the free scale parameter $\hat{\eta}$, which jointly marginalize over the unknown multiplicative term.

This scenario exemplifies a classical estimation problem in which the observations of $P_{r,i}$ for each orientation i are nonlinearly related to the unknown receiver coordinates and corrupted by AWGN of known variance. Although no unbiased efficient estimator exists for this NL model, one can derive a MVU estimate by solving a constrained NL least-squares problem that minimizes the following cost function:

$$F(x, y, z) = \sum_{i=1}^{K} F_i(x, y, z), \tag{40}$$

where:

$$F_i(x, y, z) = \sum_{k=1}^{N} \big(P_{r,i,k} + C\, Q_i^m(x, y, z)\, L(x, y, z)\big)^2. \tag{41}$$

Here $P_{r,i,k}$ denotes the k-th instantaneous received power sample at transmitter's orientation i. Moreover, the polynomial terms $Q_i(x, y, z)$ and $L(x, y, z)$ must satisfy $Q_i(x, y, z) \geq 0$ for $i = 1, \ldots, K$ and $L(x, y, z) \leq 0$, corresponding respectively to $\cos\phi_i \geq 0$ and $\cos\psi \geq 0$ for incidence and irradiance angles in $[0°, 90°]$. Thus, the NL direction-finding estimator is:

$$\hat{\mathbf{n}}_{d,\text{NL}} = \arg \min_{\substack{(x,y,z-H) \in \mathbb{S}^2 \\ Q_i(x,y,z) \geq 0,\ i=1,\ldots,K \\ L(x,y,z) \leq 0}} F(x, y, z), \tag{42}$$

where $\hat{\mathbf{n}}_{d,\text{NL}}$ denotes the estimated direction vector obtained with the NL method. In practice, this optimisation can be performed in MATLAB using the `solve` function, which internally invokes `fmincon` and employs gradient-descent algorithms suitable for constrained NL problems.

VI. Linear Estimators

The present section develops the direction-finding stage with full mathematical rigour and presents two statistically grounded estimators: (i) generalized least squares, i.e., GLS, which is minimum-variance under Gaussian noise with known covariance, and (ii) weighted least squares, i.e., WLS, a less expensive approximation that neglects the weak cross–correlation between equations. For implementation details, an end-to-end pseudocode of the full procedure (GLS/WLS direction finding, distance recovery, and 3D position estimation) is provided in Algorithm 1.

A. Direction Estimation via GLS

This subsection consolidates the analytical procedure that converts the raw optical powers into a closed-form, statistically efficient estimate of the direction vector $\mathbf{n}_d$. The exposition proceeds from the deterministic constraints, through a rigorous stochastic characterisation of the residuals, to the final GLS solution.

1) Lambertian Power Ratios and Homogeneous Constraints: Let $\mu_i = \mathbb{E}[P_{r,i}]$ be the noise-free average power at orientation i. We define power ratio β_i as:

$$\beta_i = \left(\frac{\mu_i}{\mu_1}\right)^{1/m}, \quad i = 2, \ldots, K, \tag{43}$$

and, from (4), (2), (3) and (8):

$$\mu_i = -C\,\frac{\big(\mathbf{n}_{t,i}\cdot\mathbf{d}\big)^m\big(\mathbf{n}_r\cdot\mathbf{d}\big)}{d^{m+3}}. \tag{44}$$

The distance d and the receiver-dependent factor $\cos\psi$ appear identically in both μ_i and μ_1. Their ratio therefore cancels these common factors, yielding $\beta_i = \cos\phi_i/\cos\phi_1$ from (3), which depends only on the LED orientations and the displacement direction $\mathbf{d}$.

Using (44) in (43), each β_i imposes the orthogonality condition:

$$\mathbf{a}_i\cdot\mathbf{d} = 0, \quad i = 2, \ldots, K, \tag{45}$$

i.e., $\mathbf{a}_i \perp \mathbf{d}$, with:

$$\mathbf{a}_i = \mathbf{n}_{t,i} - \beta_i\mathbf{n}_{t,1}, \tag{46}$$

which therefore supplies $K-1$ homogeneous hyperplanes that intersect at the true $\mathbf{d}$. Geometrically, each constraint $\mathbf{a}_i\cdot\mathbf{d} = 0$ defines a hyperplane through the origin containing the true direction $\mathbf{d}$. With $K-1$ such hyperplanes (from K orientations), their intersection yields the direction estimate.

2) Stochastic Perturbation of the Ratios: Recall from (5) that the sample mean is:

$$\bar{P}_{r,i} = \frac{1}{N}\sum_{k=1}^{N} P_{r,k,i} \sim \mathcal{N}(\mu_i, \sigma^2/N). \tag{47}$$

For notational convenience, define $\hat{\mu}_i \triangleq \bar{P}_{r,i}$, which is the MVU estimator of μ_i. Injecting $\hat{\mu}_i$ into (43) yields:

$$\hat{\beta}_i = g(\hat{\mu}_1, \hat{\mu}_i), \qquad g(x_1, x_2) = \left(\frac{x_2}{x_1}\right)^{\frac{1}{m}}. \tag{48}$$

As demonstrated in Appendix A, a first-order Taylor expansion of g centred at $\boldsymbol{\mu}_i = (\mu_1, \mu_i)^\mathsf{T}$ gives:

$$\hat{\beta}_i \approx \beta_i + \frac{\beta_i}{m}\left(\frac{n_i}{\mu_i} - \frac{n_1}{\mu_1}\right). \tag{49}$$

Hence every $\hat{\beta}_i$ shares the common perturbation n_1, producing correlation across orientations.

Let us now define the ratio error:

$$\tilde{n}_i = \hat{\beta}_i - \beta_i. \tag{50}$$

Then, as demonstrated in Appendix B, using (49) and the independence of n_i and n_1 leads to:

$$\mathrm{Var}[\tilde{n}_i] = \frac{\sigma^2}{Nm^2}\,\beta_i^2\big(\mu_i^{-2} + \mu_1^{-2}\big), \tag{51}$$

$$\mathrm{Cov}[\tilde{n}_i, \tilde{n}_j] = \frac{\sigma^2}{Nm^2}\,\beta_i\beta_j\,\mu_1^{-2}, \quad i \neq j. \tag{52}$$

The factor $(\mu_i^{-2} + \mu_1^{-2})$ in (51) demonstrates that the variance of $\tilde{n}_i$ depends explicitly on μ_i, even though each n_i has the same variance σ^2. In other words, a single-LED system exhibits heteroscedastic ratio errors.

3) Residuals and Their Covariance: Define the empirical constraint normals as $\hat{\mathbf{a}}_i = \mathbf{n}_{t,i} - \hat{\beta}_i\,\mathbf{n}_{t,1}$ and the corresponding residuals as:

$$\xi_i = \hat{\mathbf{a}}_i\cdot\mathbf{d}, \qquad i = 2, \ldots, K. \tag{53}$$

Using (50) and (46), we have $\hat{\mathbf{a}}_i = \mathbf{a}_i - \tilde{n}_i\,\mathbf{n}_{t,1}$. Since $\mathbf{a}_i\cdot\mathbf{d} = 0$ by (45), it follows that:

$$\xi_i = -\big(\mathbf{n}_{t,1}\cdot\mathbf{d}\big)\,\tilde{n}_i = -\,\kappa\,\tilde{n}_i, \qquad \kappa = \mathbf{n}_{t,1}\cdot\mathbf{d}. \tag{54}$$

Stacking $\boldsymbol{\xi} = [\xi_2, \ldots, \xi_K]^\mathsf{T}$ and $\tilde{\boldsymbol{n}} = [\tilde{n}_2, \ldots, \tilde{n}_K]^\mathsf{T}$ yields:

$$\begin{aligned} \boldsymbol{\xi} &= -\,\kappa\,\tilde{\boldsymbol{n}}, \\ \mathbb{E}[\boldsymbol{\xi}] &= \mathbf{0}_{1\times(K-1)}, \\ \boldsymbol{\Sigma}_{\boldsymbol{\xi}} &= \mathrm{Cov}(\boldsymbol{\xi}) = \kappa^2\boldsymbol{\Sigma}_\beta, \end{aligned} \tag{55}$$

where $\boldsymbol{\Sigma}_\beta \in \mathbb{R}^{(K-1)\times(K-1)}$ collects the variances and covariances of $\tilde{n}_i$ given in (51)–(52).

4) GLS Formulation and Closed-Form Solution: Stack the empirical constraint normals into $\hat{\mathbf{A}} = [\hat{\mathbf{a}}_2 \cdots \hat{\mathbf{a}}_K] \in \mathbb{R}^{3\times(K-1)}$. From (53)–(55), the stacked residuals satisfy $\boldsymbol{\xi} = \hat{\mathbf{A}}^\mathsf{T}\mathbf{d}$ with $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}_{1\times(K-1)}, \boldsymbol{\Sigma}_{\boldsymbol{\xi}})$ and $\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \kappa^2\boldsymbol{\Sigma}_\beta$, $\kappa = \mathbf{n}_{t,1}\cdot\mathbf{d}$. Given an observation of $\boldsymbol{\xi} = \hat{\mathbf{A}}^\mathsf{T}\mathbf{d}$, the maximum-likelihood (equivalently, GLS) estimate of the direction is obtained by minimizing the Mahalanobis norm of $\boldsymbol{\xi}$ [42]:

$$\begin{aligned} \hat{\mathbf{d}} &= \arg\min_{\|\mathbf{d}\|=1} \big\|\hat{\mathbf{A}}^\mathsf{T}\mathbf{d}\big\|^2_{\boldsymbol{\Sigma}_{\boldsymbol{\xi}}^{-1}} \\ &= \arg\min_{\|\mathbf{d}\|=1} \mathbf{d}^\mathsf{T}\big(\hat{\mathbf{A}}\,\boldsymbol{\Sigma}_{\boldsymbol{\xi}}^{-1}\hat{\mathbf{A}}^\mathsf{T}\big)\mathbf{d}. \end{aligned} \tag{56}$$

Since $\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \kappa^2\boldsymbol{\Sigma}_\beta$, the unknown scale κ^2 cancels in (56); hence one may use $\boldsymbol{\Sigma}_\beta^{-1}$ in place of $\boldsymbol{\Sigma}_{\boldsymbol{\xi}}^{-1}$ without affecting the minimizer. The problem is a constrained Rayleigh quotient whose solution is the unit eigenvector associated with the smallest eigenvalue of:

$$\mathbf{M}_{\mathrm{GLS}} = \hat{\mathbf{A}}\,\boldsymbol{\Sigma}_\beta^{-1}\hat{\mathbf{A}}^\mathsf{T} \in \mathbb{R}^{3\times 3}. \tag{57}$$

Therefore:

$$\hat{\mathbf{n}}_{d,\mathrm{GLS}} = \frac{\mathbf{v}_{\min}\big(\mathbf{M}_{\mathrm{GLS}}\big)}{\big\|\mathbf{v}_{\min}\big(\mathbf{M}_{\mathrm{GLS}}\big)\big\|}, \qquad \hat{\mathbf{n}}_{d,\mathrm{GLS}}\cdot\mathbf{n}_{t,1} > 0, \tag{58}$$

where $\mathbf{v}_{\min}(\cdot)$ denotes the eigenvector corresponding to the smallest eigenvalue; the sign convention ensures the direction points from the transmitter toward the receiver.

Algorithm 1 3D Position Estimation from K Orientations

Require: Power samples $P_{r,i,k}$ ($k = 1, \dots, N$, $i = 1, \dots, K$); steering vectors $\mathbf{n}_{t,i}$; Lambertian order m; noise variance σ^2; MODE$\in \{\mathrm{GLS}, \mathrm{WLS}\}$; radiometric constant C.
Ensure: $\hat{\mathbf{r}}$ (3D position of the receiver)

Direction finding:
  Compute $\hat{\mu}_i = \frac{1}{N}\sum_{k=1}^{N} P_{r,k,i}$ for $i = 1, \dots, K$
  Compute $\hat{\beta}_i = (\hat{\mu}_i/\hat{\mu}_1)^{1/m}$ for $i = 2, \dots, K$
  Form $\mathbf{a}_i = \mathbf{n}_{t,i} - \hat{\beta}_i\, \mathbf{n}_{t,1}$
  Set $\mathbf{A} = [\mathbf{a}_2, \dots, \mathbf{a}_K]$
if MODE = GLS **then**
  Build $\boldsymbol{\Sigma}_\beta$ via (51)–(52), substituting $\beta_i \leftarrow \hat{\beta}_i$
  $\mathbf{M} \leftarrow \mathbf{A}\,\boldsymbol{\Sigma}_\beta^{-1}\,\mathbf{A}^\mathsf{T}$
else ▷ WLS (diagonal approximation)
  $\tilde{\boldsymbol{\Sigma}}_\beta \leftarrow \mathrm{diag}\{\hat{\beta}_i^2(\hat{\mu}_i^{-2} + \hat{\mu}_1^{-2})\}$
  $\mathbf{M} \leftarrow \mathbf{A}\,\tilde{\boldsymbol{\Sigma}}_\beta^{-1}\,\mathbf{A}^\mathsf{T}$
end if
  Compute $\mathbf{v}_{\min}$ = eigenvector of M with smallest eigenvalue
  $\hat{\mathbf{n}}_d \leftarrow \mathbf{v}_{\min}/\|\mathbf{v}_{\min}\|$
if $\hat{\mathbf{n}}_d \cdot \mathbf{n}_{t,1} < 0$ **then**
  $\hat{\mathbf{n}}_d \leftarrow -\,\hat{\mathbf{n}}_d$ ▷ Ensure pointing to Rx
end if

Distance estimation (~~beamsteering~~ cooperative alignment):
  Steer LED to $\mathbf{n}_t \leftarrow \hat{\mathbf{n}}_d$, Reorient PD to $\mathbf{n}_r \leftarrow -\,\hat{\mathbf{n}}_d$
  Collect N power samples $P_{r,K+1}$
  Compute $\hat{\mu}_{K+1} = \frac{1}{N}\sum_{k=1}^{N} P_{r,K+1,k}$
  $\hat{d} \leftarrow \sqrt{\frac{C}{\hat{\mu}_{K+1}}}$

Final 3D position:
  $\hat{\mathbf{r}} \leftarrow \mathbf{t} + \hat{d}\,\hat{\mathbf{n}}_d$
return $\hat{\mathbf{r}}$

B. WLS as a Practical Simplification

The GLS solution in (56) requires inverting the $(K-1)\times(K-1)$ covariance $\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \kappa^2\boldsymbol{\Sigma}_\beta$, typically via a Cholesky factorization [47]. For real-time, low-power receivers, a lighter alternative is to neglect the off-diagonal terms in (52) and approximate:

$$\tilde{\boldsymbol{\Sigma}}_\beta = \mathrm{diag}\Big\{\beta_i^2\big(\mu_i^{-2} + \mu_1^{-2}\big)\Big\}_{i=2}^{K}. \tag{59}$$

Since $\boldsymbol{\Sigma}_{\boldsymbol{\xi}} = \kappa^2\boldsymbol{\Sigma}_\beta$, the unknown scale κ^2 cancels, and we define the WLS weighting matrix as $\tilde{\boldsymbol{\Sigma}}_\beta^{-1}$.

Using the empirical normals $\hat{\mathbf{A}}$ from the previous subsection, the WLS criterion is the Rayleigh quotient:

$$J_{\mathrm{WLS}}(\mathbf{d}) = \mathbf{d}^\mathsf{T}\big(\hat{\mathbf{A}}\,\tilde{\boldsymbol{\Sigma}}_\beta^{-1}\,\hat{\mathbf{A}}^\mathsf{T}\big)\mathbf{d}, \qquad \|\mathbf{d}\| = 1, \tag{60}$$

and we introduce the associated WLS matrix:

$$\mathbf{M}_{\mathrm{WLS}} = \hat{\mathbf{A}}\,\tilde{\boldsymbol{\Sigma}}_\beta^{-1}\,\hat{\mathbf{A}}^\mathsf{T} \in \mathbb{R}^{3\times3}. \tag{61}$$

The minimizer is the unit eigenvector corresponding to the smallest eigenvalue of $\mathbf{M}_{\mathrm{WLS}}$:

$$\hat{\mathbf{n}}_{d,\mathrm{WLS}} = \frac{\mathbf{v}_{\min}\big(\mathbf{M}_{\mathrm{WLS}}\big)}{\big\|\mathbf{v}_{\min}\big(\mathbf{M}_{\mathrm{WLS}}\big)\big\|}, \qquad \hat{\mathbf{n}}_{d,\mathrm{WLS}}\cdot\mathbf{n}_{t,1} > 0. \tag{62}$$

The neglected cross-covariances in (52) scale with μ_1^{-2}, whereas the retained diagonal terms scale with $\mu_i^{-2} + \mu_1^{-2}$. Thus, when the reference orientation $i = 1$ is chosen such that $\mu_1 \gg \mu_i$ for most i, the off-diagonal terms are negligible relative to the diagonal terms and WLS closely tracks GLS while avoiding a full covariance inversion. In more symmetric geometries where $\mu_i \approx \mu_1$ for many i, the cross-correlations are non-negligible and the full GLS should be used for optimal performance.

C. Receiver-Orientation Independence

Having defined all three direction estimators—NLS in Section V, and GLS/WLS above—we now establish a property common to all of them: the direction estimates are independent of the receiver orientation $\mathbf{n}_r$. From (22)–(24), the channel model factorizes as $\mu_i = \eta\, Q_i^m$, where $\eta = C\cos\psi/d^2$ is a positive scalar common to all K orientations, since the receiver remains stationary throughout the K-orientation acquisition.

For GLS and WLS, the power ratios $\beta_i = (\mu_i/\mu_1)^{1/m} = Q_i/Q_1$ (43) cancel η exactly. Since the constraint vectors $\mathbf{a}_i$ (46), the covariance $\boldsymbol{\Sigma}_\beta$ (51)–(52), and the matrices $\mathbf{M}_{\mathrm{GLS}}$ (57) and $\mathbf{M}_{\mathrm{WLS}}$ (61) are all constructed from $\{\beta_i\}$ and $\{\mathbf{n}_{t,i}\}$, the direction estimates are $\mathbf{n}_r$-independent. The GLS weighting via $\boldsymbol{\Sigma}_\beta^{-1}$ further ensures that hyperplanes (45) with lower noise variance contribute more to the eigenvector solution.

For NLS, the normalized targets $p_i = \mu_i/\max_j \mu_j = Q_i^m/Q_{\max}^m$ (36) cancel η through a different mechanism: the free scale parameter $\hat{\eta}$ absorbs the unknown common factor during optimization (cf. Section V).

This property holds regardless of whether $\mathbf{n}_r$ is known or unknown, and numerical confirmation under random receiver tilts is reported in Section VII.

VII. Estimators Comparison

A. Direction-Finding Performance

This subsection evaluates the direction-finding stage in isolation, using the angular error $\theta_{\mathrm{err}} = \arccos(\hat{\mathbf{n}}_d \cdot \mathbf{n}_d)$ as the performance metric, independently of distance recovery. For each transmitter orientation i, the received optical power was simulated according to (4) and (2) using the system parameters in Table II, with a semi-angle at half power $\Phi_{1/2} = 45°$. Zero-mean AWGN $n_i \sim \mathcal{N}(0, \sigma^2)$ was drawn i.i.d. across orientations and samples; the sample mean $\bar{P}_{r,i}$ was formed from N realizations as in (5). For each estimator, $M = 1{,}000$ Monte Carlo trials were performed at each of the $|\mathcal{R}| = 1{,}792$ testbed positions for $K = 5$ orientations. To ensure a fair comparison, each estimator uses its best GA-optimized orientation set: GLS and WLS use the set optimized with the GLS angular RMSE as cost function, while NLS uses the set optimized with the DEB as cost function.

1) Angular Error CDF: Table IV and Fig. 6 (solid curves) report the spatial RMSE, $\mathrm{CDF}_{90\%}$, and mean angular error. The hierarchy DEB $<$ NLS $<$ GLS $<$ WLS is consistent with theory: the DEB (0.52°) is the Cramér–Rao bound; NLS (0.63°) operates on the full nonlinear Lambertian model without linearization; GLS (0.66°) linearizes the ratio model but uses the optimal $\boldsymbol{\Sigma}_\beta^{-1}$ weighting; and WLS (0.77°) further simplifies to diagonal weights. All estimators achieve sub-degree RMSE for direction finding.

TABLE IV
DIRECTION-FINDING ANGULAR ERROR ($K = 5$, $M = 1{,}000$ TRIALS).

Method	RMSE [°]	CDF 90% [°]	Mean [°]
DEB	0.52	0.75	0.46
NLS	0.63	0.81	0.52
GLS	0.66	0.99	0.54
WLS	0.77	1.14	0.61

TABLE V
DIRECTION-FINDING RMSE UNDER RANDOM RECEIVER TILT ($K = 5$, HALF-NORMAL σ=5°, max =30°).

Method	Baseline [°]	Random tilt [°]	Degradation
GLS	0.657	0.671	+2.24%
WLS	0.766	0.786	+2.62%
NLS	0.620	0.631	+1.68%
DEB	0.518	0.527	+1.69%

2) Robustness to Receiver Tilt: To numerically validate the $\mathbf{n}_r$-independence established in Section VI-C, we evaluate direction-finding performance under random receiver tilts. Each tilted receiver normal is parameterized as $\mathbf{n}_r = [\sin\theta_{\text{tilt}}\cos\varphi_{\text{tilt}},\ \sin\theta_{\text{tilt}}\sin\varphi_{\text{tilt}},\ \cos\theta_{\text{tilt}}]^\mathsf{T}$, where θ_{tilt} is the polar angle between $\mathbf{n}_r$ and the vertical axis $\mathbf{u}_z$ and φ_{tilt} is the azimuth in the horizontal plane. The tilt angle θ_{tilt} is drawn from a half-normal distribution with $\sigma = 5°$ truncated at $30°$, and φ_{tilt} is uniform on $[0°, 360°)$. For each of the $1{,}792$ positions, $N_{\text{tilt}} = 100$ independent random tilts are generated, with $M = 1{,}000$ noise trials per tilt realization.

Table V reports the RMSE degradation relative to the vertical baseline ($\mathbf{n}_r = [0, 0, 1]^\mathsf{T}$). All estimators exhibit degradation below 3%, confirming that receiver-orientation variations have negligible impact on direction-finding accuracy. This residual degradation is not intrinsic to the estimators but is attributable to SNR reduction—a tilted PD intercepts less optical power ($\cos\psi < 1$), lowering the effective SNR—rather than structural bias. Figure 6 corroborates these results: the empirical CDFs of the per-position angular RMSE under random tilt (dashed curves) are virtually indistinguishable from those obtained with a vertical receiver (solid curves) for all four methods, with the inset depicting the truncated half-normal tilt distribution applied in the simulation.

B. *Positioning Performance 3D Positioning Performance*

This section compares the model-based estimators introduced earlier (NL, WLS, and GLS). For each transmitter orientation i, the received optical power was simulated according to (4) and (2) using the system parameters in Table II, with a semi-angle at half power $\Phi_{1/2} = 45°$. Zero-mean AWGN $n_i \sim \mathcal{N}(0, \sigma^2)$ was drawn i.i.d. across orientations and samples; the sample mean $\bar{P}_{r,i}$ was formed from N realizations as in (5). Position estimates were then computed over the 3D testbed specified in Table II, using the GA-optimized orientation sets in Table III. Estimates were compared against the ground-truth positions, and the 3D Euclidean position error was recorded for each trial. Figure 7 plots the cumulative distribution function (CDF) of the 3D position error for $K \in \{3, 5, 9\}$. The case $K = 3$ uses the deterministic baseline of [22], which was originally proposed for 2D estimation; however, by applying

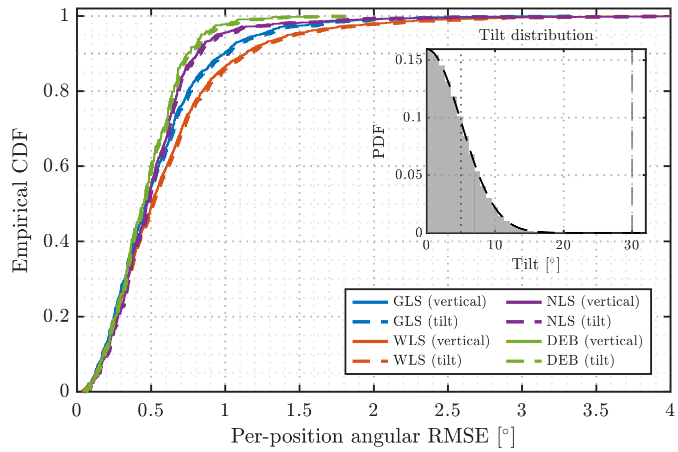

Fig. 6. Empirical CDF of the per-position angular RMSE for K=5 orientations (M=1,000 noise trials, 1,792 positions). Solid curves: vertical receiver ($\mathbf{n}_r$=$[0, 0, 1]^\mathsf{T}$); dashed curves: random receiver tilt (N_{tilt}=100 realizations per position). Each estimator uses its GA-optimized orientation set. Inset: truncated half-normal tilt distribution (σ=5°, $\theta_{\max}$=30°). The near-perfect overlap between solid and dashed curves confirms that receiver-orientation variations have negligible impact on direction-finding accuracy (Section VI-C).

the singular value decomposition (SVD) to the matrix obtained by stacking the vectors in (46), we found it can be extended to 3D, but only for three orientations (i.e., without WLS/GLS weighting). The setting $K = 5$ is selected as a latency–accuracy compromise based on the CRLB analysis, and $K = 9$ represents the upper bound of the orientation counts studied in this paper. For $K = 3$, $K = 5$ (GLS, WLS), and $K = 9$ (GLS, WLS), the GA-optimized orientation sets from Table III were used.

The NL estimator exhibited degraded performance when paired with those K-optimal sets because it minimizes (37) without an explicit weighting matrix, thereby giving comparable influence to orientations with negligible received power (e.g., $\phi_i > 90°$), which impairs convergence. To provide a fair comparison, a separate GA optimization was conducted using the parameters in Table II but with the NL method's RMSE as the objective; this NL-specific optimal set is used in the NL curves reported.

Across all estimators, the $K = 9$ configurations (dashed traces) stochastically dominate their $K = 5$ counterparts (solid lines), in agreement with the CRLB trend that additional orientations reduce the PEB. Among the model-based schemes, GLS consistently tracks the theoretical PEB most closely, regardless of K, with WLS incurring a small but visible loss due to its diagonal covariance approximation. All proposed estimators substantially outperform the deterministic $K = 3$ baseline [22]. This subsection extends the evaluation to the full two-stage pipeline (direction finding followed by distance recovery), using the same simulation setup described in Section VII-A. Position estimates were computed over the 3D testbed specified in Table II, using the GA-optimized orientation sets detailed below. As in the direction-finding evaluation (Section VII-A), each estimator uses its best orientation set: GLS and WLS employ the GLS-DF-optimized set, while NLS uses the DEB-optimized set; the PEB is

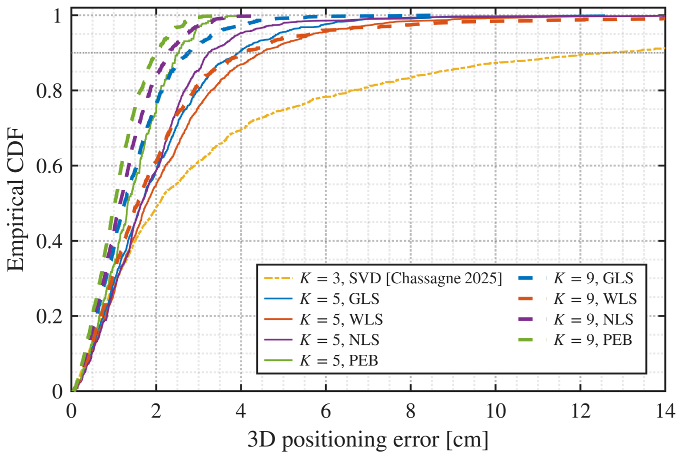

Fig. 7. Comparison of the CDFs of different 3D position estimation methods for $K = 3, 5, 9$.Empirical CDF of the 3D positioning error for $K \in \{3, 5, 9\}$ (M=1,000 Monte Carlo trials, 1,792 testbed positions). Solid lines: K=5; dashed lines: K=9; dash-dotted: K=3 SVD baseline [22]. Each estimator uses its respective GA-optimized orientation set.

also evaluated with the DEB-optimized orientations. The 3D Euclidean position error was recorded for each of M=1,000 trials at each of the $|\mathcal{R}|$=1,792 testbed positions. Figure 7 plots the empirical CDF of the per-position 3D positioning error for $K \in \{3, 5, 9\}$. The case K=3 employs the deterministic SVD baseline of [22], originally proposed for 2D estimation; by applying the SVD to the matrix formed by the vectors in (46), we extend it to 3D, albeit restricted to three orientations and without a weighted formulation. The setting K=5 is selected as a latency–accuracy compromise based on the PEB analysis, and K=9 represents the upper bound of the orientation counts studied herein. For GLS, NLS, and the PEB, the K=9 configurations (dashed traces) stochastically dominate their K=5 counterparts (solid lines), in agreement with the PEB trend that additional orientations reduce the positioning bound. All proposed estimators substantially outperform the K=3 baseline [22].

Table VI summarizes the quantitative metrics: RMSE, the 90th percentile of the error CDF ($\mathrm{CDF}_{90\%}$), and APE. For $K = 5$, GLS achieved RMSE $=$ 2.72 cm and $\mathrm{CDF}_{90\%}$ $=$ 3.86 cm, within a factor $\approx 1.75\times$ of the CRLB benchmark (1.55 cm RMSE). WLS trailed GLS by $\approx 36\%$ in RMSE at $K = 5$, reflecting the benefit of modeling inter-orientation error correlations. The NL estimator, after re-optimizing its orientation set, became competitive with WLS at $K = 5$ and surpassed WLS at $K = 9$, where NL attained an RMSE of 2.36 cm versus 2.96 cm for WLS. Increasing K from 5 to 9 delivered consistent gains for all methods (e.g., GLS RMSE improved from 2.72 cm to 2.41 cm), and the gap to the CRLB narrowed (from 1.55 cm to 1.21 cm). Overall, GLS provided the best fidelity to the bound with modest complexity; WLS offered a favorable accuracy–complexity compromise; and NL benefited markedly from orientation-set co-design, especially at larger K. For K=5, NLS achieved the lowest estimator RMSE of 2.40 cm and $\mathrm{CDF}_{90\%}$=3.23 cm, followed by GLS (2.52 cm) and WLS (2.92 cm). GLS remained within a factor $\approx 1.54\times$ of the PEB (1.64 cm RMSE). Increasing to K=9

TABLE VI
PERFORMANCE OF POSITION-ESTIMATION METHODS IN A $3 \times 3 \times 2\,\mathrm{m}^3$ ROOM.

K	Method	RMSE [cm]	CDF 90% [cm]	APE [cm]
3	[22]	11.05	12.61	5.35
5	GLS	2.52	3.93	**2.00**
5	WLS	2.92	4.49	2.25
5	NLS	2.40	3.23	1.90
5	PEB	1.64	2.53	1.43
9	GLS	1.78	2.75	**1.46**
9	WLS	3.42	4.01	2.15
9	NLS	1.47	2.28	1.26
9	PEB	1.28	2.00	1.10

Note: For the PEB rows, "RMSE" denotes the root-mean-square of the PEB over all testbed positions (a deterministic spatial statistic, not a Monte Carlo estimate); "CDF 90%" is the 90th percentile of the PEB distribution across positions; "APE" is the spatial mean PEB.

delivered gains for GLS and NLS: GLS improved to 1.78 cm (a 29% reduction) and NLS to 1.47 cm (a 39% reduction), narrowing the gap to the PEB (1.28 cm). WLS, however, degraded from 2.92 cm at K=5 to 3.42 cm at K=9; this behavior is consistent with its diagonal covariance approximation becoming increasingly suboptimal as the number of orientations grows and the off-diagonal terms neglected in the WLS simplification become non-negligible. However, NL does not surpass GLS, despite appearing more general, NL is unweighted and does not account for the relative contribution of each constrain equation in the position solve, and therefore it fails to outperform the weighted GLS method. The performance gap between the estimators and the PEB arises from two factors: (i) the two-stage architecture discards distance information contained in the absolute power levels during direction finding, whereas the PEB assumes joint estimation from all $K+1$ measurements; and (ii) the PEB assumes perfect beam alignment in the distance-recovery measurement, which no practical estimator can achieve. The APE values track the same ordering as RMSE and $\mathrm{CDF}_{90\%}$, indicating that the improvements are uniform across the distribution rather than confined to the median or the tail.

Figure 8 illustrates the 3D position estimates produced by GLS and WLS for $K = 5$ at three receiver heights $z \in \{0, 0.6, 1.2\}\,\mathrm{m}$, overlaid on the ground truth. At $z = 1.2\,\mathrm{m}$, errors increase near the testbed boundaries—particularly at the corners—consistent with the PEB heat maps, shown in Fig. 3(a), and attributable to reduced SNR at extreme incidence angles. At $z = 0\,\mathrm{m}$, the errors are more spatially uniform. In both cases, GLS maintains a tighter scatter than WLS, in line with its closer adherence to the CRLB. Figure 8 depicts the 3D position estimates produced by GLS for K=5 at three receiver heights $z \in \{0, 0.6, 1.2\}\,\mathrm{m}$, with the LED transmitter located at the ceiling ($z = 2\,\mathrm{m}$). The displacement between each reference position and the corresponding estimate reveals the spatial structure of the positioning error. At $z = 1.2\,\mathrm{m}$, errors grow towards the testbed boundaries—particularly at the corners—consistent with the DEB heat maps in Fig. 3(a) and attributable to the reduced SNR at large incidence angles. At lower heights ($z = 0$ and $z = 0.6\,\mathrm{m}$), the estimates remain tightly concentrated around the reference positions, in agreement with the low RMSE reported in Table VI.

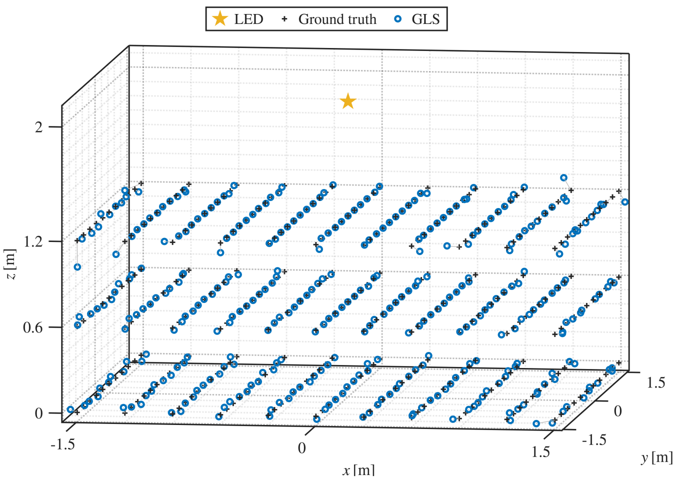

Fig. 8. 3D position estimates at three receiver heights for $K = 5$ orientations: GLS, WLS, ground truth.3D position estimates produced by GLS (K=5) at receiver heights $z \in \{0, 0.6, 1.2\}$ m in the $3 \times 3 \times 2\,\mathrm{m}^3$ room. The LED transmitter is at the ceiling ($z = 2\,\mathrm{m}$); segments connect each reference position to the corresponding estimate.

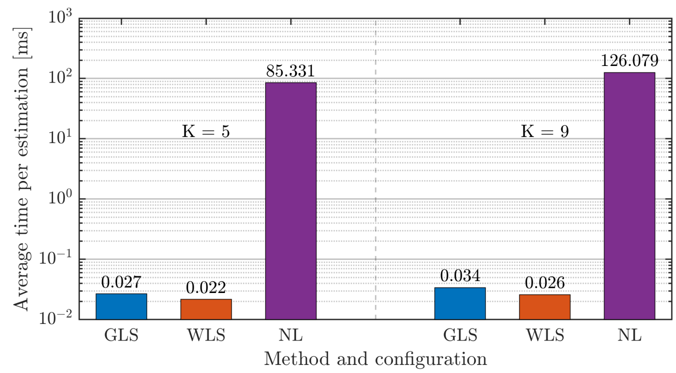

Fig. 9. Computation latency of the NLNLS, GLS, and WLS estimators for $K = 5$ and $K = 9$.

C. Computational Complexity

We summarize the per-inference computational cost for direction finding under K LED orientations and N samples per orientation; computing the sample means $\{\hat{\mu}_i\}_{i=1}^{K}$ via (5) incurs $\mathcal{O}(NK)$ and is common to all schemes. We next detail the direction–finding estimators (GLS/WLS/NLNLS); the subsequent distance–recovery step is identical across methods and requires constant time, as given by (9).

1) GLS: From (51)–(52) we assemble the full covariance $\boldsymbol{\Sigma}_\beta \in \mathbb{R}^{(K-1)\times(K-1)}$ (cost $\mathcal{O}(K^2)$), then factor it (Cholesky) and apply the solves needed to build $\mathbf{M}_{\text{GLS}} = \hat{\mathbf{A}}\,\boldsymbol{\Sigma}_\beta^{-1}\hat{\mathbf{A}}^\top$ in (57). The factorization dominates with $\mathcal{O}((K-1)^3)$ flops; the subsequent triangular solves are $\mathcal{O}(K^2)$. The eigenproblem is 3×3 and thus constant-cost. Overall, the cost per position estimate is $\mathcal{O}(NK) + \mathcal{O}(K^2) + \mathcal{O}(K^3)$ with GLS.

2) WLS: Replacing $\boldsymbol{\Sigma}_\beta$ by the diagonal $\tilde{\boldsymbol{\Sigma}}_\beta$ in (59) removes the full covariance inversion. Forming $\mathbf{M}_{\text{WLS}} = \hat{\mathbf{A}}\,\tilde{\boldsymbol{\Sigma}}_\beta^{-1}\hat{\mathbf{A}}^\top$ requires only column scalings and a $3 \times (K-1)$–by–$(K-1)\times 3$ product, i.e., $\mathcal{O}(K)$. The final eigenproblem is again constant-cost. Hence, the cost per position estimate is $\mathcal{O}(NK) + \mathcal{O}(K)$ with WLS.

3) NLNLS: Solving (37) with MATLAB's `fmincon` (interior-point) entails, at each iteration, evaluating residuals and Jacobians for K orientations and solving a constant-size normal equation; the latter is constant-cost, while Jacobian/residual assembly is $\mathcal{O}(K)$. Using the Gauss–Newton least-squares structure for cost accounting, and with T_{NL} interior-point iterations, the per-estimate cost is $\mathcal{O}(NK) + \mathcal{O}(T_{\text{NL}}K)$. Solving (37) with MATLAB's `fmincon` (interior-point) optimizes over the direction $\mathbf{v} \in \mathbb{S}^2$ and scale $\hat{\eta}$ (a constant-size parameter vector) subject to K inequality constraints. At each iteration, the solver evaluates the cost gradient and constraint Jacobian, both requiring $\mathcal{O}(K)$ operations (one term per orientation in (38)), and solves a constant-size linear system for the Newton step. With T_{NLS} interior-point iterations, the per-estimate cost is $\mathcal{O}(NK) + \mathcal{O}(T_{\text{NLS}}K)$.

Figure 9 reports the average end-to-end computation time per 3D estimate (powers pre-acquired). For K=5, WLS and GLS complete in 0.022 ms and 0.027 ms, respectively, while NLNLS requires 85.331 ms; for K=9, WLS and GLS remain in the tens of microseconds (0.026 ms and 0.034 ms), whereas NLNLS increases to 126.079 ms. The growth from K=5 to K=9 is modest for WLS (+18%) and GLS (+26%), consistent with their $\mathcal{O}(K)$ and small $\mathcal{O}(K^3)$ components, noting that the cubic term in GLS acts on a compact $(K-1)\times(K-1)$ system followed by a constant-cost 3×3 eigensolve. By contrast, NLNLS is two to three orders of magnitude slower and increases by $\approx 48\%$ with K due to the constrained non-linear solve performed with MATLAB's `fmincon` (interior-point), which repeatedly evaluates K-sized Jacobians as part of the iteration. These measurements corroborate the complexity analysis above: closed-form WLS/GLS enable real-time inference without fingerprinting or model retraining, whereas NLNLS is latency-limited.

VIII. Conclusion and Future Works

This paper introduced a beam–steered single-LED, single-PD OWP architecture that delivers full 3D indoor localization without receiver rotation, cameras, or PD arrays. Our method is explicitly model-based, grounded in radiometry and geometry, so it requires no fingerprinting or data retraining, enabling interpretable performance, principled uncertainty analysis, and efficient co-design of hardware and algorithms.

We proposed a two-stage process: (i) direction finding from K steered orientations, followed by (ii) distance recovery from a single beam-aligned power measurement. A full CRLB/PEB DEB and PEB analysis characterized identifiability ($K \geq 3$) and showed the benefits of overdetermination ($K \geq 4$). A GA-based orientation-set optimizer minimized the RMSE PEB DEB across a 3D testbed and yielded practical steering patterns. Additionally, we introduce a ratio-based linearization of the Lambertian power model that yields closed-form GLS/WLS direction estimators for stage (i). We showed (Section VI-C) that all three proposed direction estimators (GLS, WLS, NLS) are mathematically independent of the receiver orientation $\mathbf{n}_r$, which eliminates the need for receiver pose calibration during the direction-finding stage.

Simulations demonstrated centimeter-level accuracy with modest K. Optimized orientation sets achieved sub–1.5 cm median PEB across $K \in [3, 9]$, reaching $0.74\,\mathrm{cm}$ at $K = 9$. With $K = 5$, GLS attained RMSE $2.72\,\mathrm{cm}$ and $\mathrm{CDF}_{90\%} = 3.86\,\mathrm{cm}$, tracking the CRLB within a factor $\approx 1.75\times$; increasing to $K = 9$ improved GLS to RMSE $2.41\,\mathrm{cm}$. Across configurations, the proposed estimators substantially outperformed the deterministic $K = 3$ baseline [22]. Importantly, GLS/WLS reduce inference to a single 3×3 eigenproblem, yielding μs-level latency, while the NL estimator exhibits ms-level runtimes. These results confirm that closed-form, model-based estimation can closely approach the theoretical bound at real-time speeds. Monte Carlo simulations over 1,792 testbed positions with 1,000 trials confirmed the effectiveness of the proposed framework. For direction finding with $K{=}5$, all estimators achieved sub-degree angular RMSE: NLS $0.63°$, GLS $0.66°$, WLS $0.77°$, with the DEB at $0.52°$. For full 3D positioning, NLS attained RMSE $2.40\,\mathrm{cm}$ and GLS $2.52\,\mathrm{cm}$ at $K{=}5$, tracking the PEB ($1.64\,\mathrm{cm}$) within a factor $\approx 1.5\times$; increasing to $K{=}9$ improved NLS to $1.47\,\mathrm{cm}$ and GLS to $1.78\,\mathrm{cm}$. All proposed estimators substantially outperformed the deterministic $K{=}3$ baseline [22] ($11.05\,\mathrm{cm}$). Importantly, GLS/WLS reduce inference to a single 3×3 eigenproblem, yielding μs-level latency ($\approx 0.03\,\mathrm{ms}$), while NLS requires $\approx 85\,\mathrm{ms}$ due to its iterative constrained optimization. Furthermore, under random receiver tilts (half-normal, $\sigma{=}5°$, $\max{=}30°$), all estimators degrade by less than 3%, confirming that receiver-orientation variations have negligible effect on direction-finding accuracy, as established analytically in Section VI-C.

Beyond accuracy, the architecture is deployment-friendly. By concentrating complexity in the infrastructure (beam steering at a single NIR source) and using only one PD per receiver, the system avoids dense LED constellations and specialized user hardware. In operation, the transmitter iteratively cycles through the predefined set of K steering orientations; this periodic scan can be exploited by multiple receivers present in the scene, enabling concurrent localization for pedestrians and other dense multi-receiver scenarios (e.g., robot swarms) with minimal on-robot implementation. Beyond accuracy, direction finding constitutes the core enabling stage of the architecture. It is fully passive on the receiver side (the PD remains at an arbitrary fixed orientation during the K measurements), requires no receiver pose calibration, and the same periodic K-orientation scan can serve multiple receivers simultaneously. Once the direction $\hat{\mathbf{n}}_d$ is known, it enables cooperative beam alignment—steering the LED toward the PD and reorienting the PD toward the LED—which maximizes the link SNR for a subsequent distance-recovery measurement that completes the 3D position. Thus, beam-steering complexity is concentrated in the infrastructure (a single source), while the receiver requires only a single PD and one cooperative reorientation for the ranging step. The full pipeline acquires $K + 1$ sequential measurements (K for direction finding and one for distance recovery). For $K = 5$ with state-of-the-art steering mechanisms, the total acquisition window is on the order of milliseconds, resulting in sub-centimeter displacement at pedestrian speeds.

Possible future work includes augmenting the radiometric–geometric model with physics-informed neural networks to learn small residual corrections while preserving physical constraints, enabling a gray-box (rather than black-box) estimator. Future work also includes validating the approach experimentally and extending the framework to OWC, with emphasis on single-source optical integrated sensing and communication. Future work includes: (i) experimental validation; (ii) integration with beam-steered OWC links, where the direction-finding stage can be embedded into the beam-tracking loop and an intelligent refresh strategy can exploit the previous direction estimate to reduce the number of required orientations; and (iii) extending the framework to optical integrated sensing and communication (ISAC).

Appendix A
Proof of Equation (49)

For each orientation i, the sample mean $\hat{\mu}_i \triangleq \bar{P}_{r,i}$ satisfies $\hat{\mu}_i = \mu_i + n_i$ with $n_i \sim \mathcal{N}(0, \sigma^2/N)$ mutually independent [43]. For $i \geq 2$, define $g(x_1, x_2) = (x_2/x_1)^{1/m}$ so that $\hat{\beta}_i = g(\hat{\mu}_1, \hat{\mu}_i)$ and $\beta_i = g(\mu_1, \mu_i)$. A first–order multivariate Taylor expansion (delta method) of g at (μ_1, μ_i) gives:

$$\hat{\beta}_i \;\approx\; \beta_i + \frac{\partial g}{\partial x_1}(\mu_1, \mu_i)\, n_1 + \frac{\partial g}{\partial x_2}(\mu_1, \mu_i)\, n_i. \tag{63}$$

The partial derivatives of g are, for general (x_1, x_2):

$$\frac{\partial g}{\partial x_1}(x_1, x_2) \;=\; -\frac{1}{m}\, x_2^{\frac{1}{m}}\, x_1^{-\frac{1}{m}-1} \;=\; -\frac{g(x_1, x_2)}{m\, x_1}, \tag{64a}$$

$$\frac{\partial g}{\partial x_2}(x_1, x_2) \;=\; \frac{1}{m}\, x_2^{\frac{1}{m}-1}\, x_1^{-\frac{1}{m}} \;=\; \frac{g(x_1, x_2)}{m\, x_2}. \tag{64b}$$

Evaluating at (μ_1, μ_i) (so $g(\mu_1, \mu_i) = \beta_i$) gives:

$$\frac{\partial g}{\partial x_1}(\mu_1, \mu_i) \;=\; -\frac{\beta_i}{m\, \mu_1}, \tag{65a}$$

$$\frac{\partial g}{\partial x_2}(\mu_1, \mu_i) \;=\; \frac{\beta_i}{m\, \mu_i}. \tag{65b}$$

Hence:

$$\hat{\beta}_i \;\approx\; \beta_i + \frac{\beta_i}{m}\left(\frac{n_i}{\mu_i} - \frac{n_1}{\mu_1}\right), \tag{66}$$

which is (49). □

Appendix B
Proof of Equation (52)

From the first–order expansion (delta method):

$$\hat{\beta}_i \;\approx\; \beta_i + \frac{\beta_i}{m}\left(\frac{n_i}{\mu_i} - \frac{n_1}{\mu_1}\right), \quad n_\ell \sim \mathcal{N}\left(0, \tfrac{\sigma^2}{N}\right) \text{ i.i.d.}, \tag{67}$$

we can deduce that the ratio error is:

$$\tilde{n}_i \;\triangleq\; \hat{\beta}_i - \beta_i \;\approx\; \frac{\beta_i}{m}\left(\frac{n_i}{\mu_i} - \frac{n_1}{\mu_1}\right). \tag{68}$$

Using $\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y)$ for independent X, Y:

$$\mathrm{Var}[\tilde{n}_i] = \left(\tfrac{\beta_i}{m}\right)^2\left(\tfrac{\sigma^2}{N\,\mu_i^2} + \tfrac{\sigma^2}{N\,\mu_1^2}\right) = \frac{\sigma^2}{Nm^2}\,\beta_i^2\left(\mu_i^{-2} + \mu_1^{-2}\right), \tag{69}$$

which is (51). For $i \neq j$:

$$\begin{aligned}\mathrm{Cov}[\tilde{n}_i, \tilde{n}_j] &= \left(\tfrac{\beta_i}{m}\right)\left(\tfrac{\beta_j}{m}\right)\mathrm{Cov}\left(\frac{n_i}{\mu_i} - \frac{n_1}{\mu_1}, \frac{n_j}{\mu_j} - \frac{n_1}{\mu_1}\right) \\ &= \left(\tfrac{\beta_i\beta_j}{m^2}\right)\left[0 - 0 - 0 + \mathrm{Var}\left(\tfrac{n_1}{\mu_1}\right)\right] \\ &= \frac{\sigma^2}{Nm^2}\,\beta_i\beta_j\,\mu_1^{-2}, \end{aligned} \tag{70}$$

since n_i, n_j, n_1 are mutually independent and $\mathrm{Var}(n_1/\mu_1) = \sigma^2/(N\mu_1^2)$. Thus:

$$\mathrm{Cov}[\tilde{n}_i, \tilde{n}_j] = \frac{\sigma^2}{Nm^2}\,\beta_i\beta_j\,\mu_1^{-2}, \quad i \neq j, \tag{71}$$

which is (52). □

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